



BIO 004 · HUMAN ANATOMY · SOLANO COLLEGE

# Module 1 Packet.

Introduction to Anatomy · Cell Anatomy · Histology · Integumentary

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## PART A

### Course Notes

Five chapters, in the order the module teaches them. Your reference for every video, lab, and exam question in Module 1.

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## PART B

### Pre-work

Four packets: Intro to Anatomy, Cell Anatomy, Tissues and Histology, Integumentary. Complete these by hand before the class day they belong to. These pages print landscape.

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## PART C

### Lab Structure List

The Module 1 slice of the master lab list. Everything here is testable on the lab portion of Exam 1.

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# Contents

Five concepts, in the order the module teaches them. Part A of the Module 1 packet. Part B is the pre-work, Part C is the lab structure list.

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INTRODUCTION TO ANATOMY

## 2 Body Cavities & Regions

INTRODUCTION TO ANATOMY

## 3 Anatomy of the Cell

CELL ANATOMY

## 4 Histology: The Four Tissue Types

TISSUES & HISTOLOGY

## 5 The Integumentary System

INTEGUMENTARY & SKIN



### Everything else for Module 1

Scan for the course home: the concept videos, Mastery OS, the calendar, and the exam schedule. Nothing behind this code is reprinted in this packet.

### WHAT MODULE 1 COVERS

Module 1 runs across the first three weeks of the term and ends in Exam 1. It carries 23 competencies, listed under each section title by their Mastery OS identifiers.

The five concepts build on one another in order. The language of anatomy gives you the words. The body cavities give you the map. The cell is the unit every tissue is made from. Histology is the four tissues those cells build. The integumentary system is the first organ where all four appear at once, so it is where the module's ideas get used rather than listed.

This is anatomy. Where function appears in these notes it is marked as a physiology note, and it is there because the structure does not make sense without it.

PART A

# Course Notes.

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Five chapters, in the order the module teaches them. Read a chapter before the class day it belongs to, then do the matching pre-work packet in Part B. This is your reference for every video, lab, and exam question in Module 1.

**1**    **The Language of Anatomy**

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**2**    **Body Cavities & Regions**

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**3**    **Anatomy of the Cell**

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**4**    **Histology: The Four Tissue Types**

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**5**    **The Integumentary System**

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## SECTION 1

## The Language of Anatomy

## INTRODUCTION TO ANATOMY

The shared vocabulary the rest of the course is built on. Every position, plane, direction, and region named here gets used in lab, on exams, and in every clinical description you will ever read.

**COMPETENCIES** W1-LEVELS-ORGANIZATION, W1-ANATOMICAL-POSITION, W1-PLANES-SECTIONS, W1-DIRECTIONAL-TERMS, W1-REGIONAL-TERMS

## BY THE END

1. Define anatomy and physiology and name the branches of each.
2. List the six levels of structural organization in order and give an example of each.
3. Name the eleven organ systems and the six basic life processes.
4. Place the body in anatomical position and explain why it is the reference for every directional term.
5. Name the planes and sections and use the paired directional terms precisely.
6. Identify the major anterior and posterior regional terms.
7. Separate disease from disorder and signs from symptoms.



WATCH AFTER YOU READ THIS CHAPTER

**Concept videos: Week 1**

[drsrennie-stack.github.io/new-build-bio4-solano/week1-concept-videos.html](https://drsrennie-stack.github.io/new-build-bio4-solano/week1-concept-videos.html)

## Anatomy and Physiology

**Anatomy** is the study of the structures in the body. **Physiology** is the study of how those structures function, how they work. The two are inseparable in practice. Structure sets the limits on what a part can do, and function explains why a structure is shaped the way it is. This course is the anatomy half.

## Branches of anatomy

- **Gross (macroscopic) anatomy**, structures visible to the unaided eye. Studied **regionally** (all structures in one body area), **systemically** (one organ system at a time), or as **surface anatomy** (internal structures located by external landmarks).
- **Microscopic anatomy**, structures too small to see unaided. Includes **cytology** (cells) and **histology** (tissues).
- **Developmental anatomy**, structural change across the lifespan. **Embryology** is the subdivision covering development before birth.
- **Pathological anatomy**, structural change caused by disease.
- **Radiographic anatomy**, internal structure seen through imaging.

## Branches of physiology

- **Neurophysiology**, function of the nervous system.
- **Endocrinology**, hormones and how they regulate function.
- **Cardiovascular physiology**, function of the heart and vessels.
- **Respiratory physiology**, function of the airways and lungs.
- **Renal physiology**, function of the kidneys.
- **Immunology**, how the body defends itself.
- **Pathophysiology**, functional change caused by disease.

## Levels of Structural Organization

The body is organized from the simplest level to the most complex. Each level is built from the one before it, so a failure at a lower level always has consequences above it.

| THE SIX LEVELS, LEAST TO MOST COMPLEX |                                                                     |                                |
|---------------------------------------|---------------------------------------------------------------------|--------------------------------|
| Level                                 | What it is                                                          | Example                        |
| Chemical                              | Atoms and molecules, the smallest level                             | Oxygen, water, a protein       |
| Cellular                              | The smallest living unit of the body                                | A keratinocyte, a neuron       |
| Tissue                                | Groups of similar cells performing a specific function              | Stratified squamous epithelium |
| Organ                                 | A structure composed of two or more tissue types with a defined job | The skin, the stomach          |
| Organ system                          | Organs working together toward one broad function                   | The integumentary system       |
| Organismal                            | All organ systems together, the entire living being                 | The person                     |

## The Eleven Organ Systems

| SYSTEM, MAJOR COMPONENTS, PRIMARY FUNCTION |                                                        |                                                                              |
|--------------------------------------------|--------------------------------------------------------|------------------------------------------------------------------------------|
| System                                     | Major components                                       | Primary function                                                             |
| Integumentary                              | Skin, hair, nails, cutaneous glands                    | Covers and protects, regulates temperature, senses the environment           |
| Skeletal                                   | Bones, cartilage, ligaments, joints                    | Supports, protects, stores minerals, houses blood formation                  |
| Muscular                                   | Skeletal muscles and their tendons                     | Produces movement, maintains posture, generates heat                         |
| Nervous                                    | Brain, spinal cord, nerves, sensory receptors          | Fast control, detects and responds to change                                 |
| Endocrine                                  | Pituitary, thyroid, adrenals, pancreas, gonads         | Slow control through hormones                                                |
| Cardiovascular                             | Heart, blood vessels, blood                            | Transports oxygen, nutrients, wastes, and heat                               |
| Lymphatic and immune                       | Lymph vessels and nodes, spleen, thymus, tonsils       | Returns fluid, filters lymph, mounts immune defense                          |
| Respiratory                                | Nasal cavity, pharynx, larynx, trachea, bronchi, lungs | Gas exchange between air and blood                                           |
| Digestive                                  | Alimentary canal plus liver, gallbladder, pancreas     | Breaks down food, absorbs nutrients, eliminates residue                      |
| Urinary                                    | Kidneys, ureters, bladder, urethra                     | Filters blood, eliminates nitrogenous waste, balances water and electrolytes |
| Reproductive                               | Gonads, ducts, accessory glands, external genitalia    | Produces gametes and offspring                                               |

## The Six Basic Life Processes

These distinguish living things from non-living things. Every one of them shows up again in a later unit.

1. **Metabolism**, the sum of all chemical reactions in the body. **Catabolism** breaks molecules down, **anabolism** builds them up.
2. **Responsiveness**, the ability to detect and react to a change in the internal or external environment.
3. **Movement**, motion of the whole body, of individual organs, of single cells, or of structures inside a cell.
4. **Growth**, an increase in size, from more cells, larger cells, or more material between cells.
5. **Differentiation**, the change of an unspecialized cell into a specialized one. Stem cells become keratinocytes, neurons, muscle fibers.
6. **Reproduction**, either the formation of new cells for growth and repair, or the production of a new individual.

## Anatomical Position

**Anatomical position** is the standard reference posture: the body erect, feet flat and slightly apart, head and eyes facing forward, arms at the sides, palms facing forward.

**Why it matters.** Every directional term assumes this position. That way a description does not change with how the person is actually standing, sitting, or lying. Right and left are always the patient's right and left, not yours.

Two related terms describe a body lying down. **Supine** is lying face up, on the back. **Prone** is lying face down, on the front.

## Body Planes and Sections

A **plane** is an imaginary flat surface passing through the body. The **section** is the cut surface that plane produces. The **midline** is the imaginary vertical line running from the top of the head, through the nose, down the spine, to the pelvis. It divides the body into equal left and right halves and serves as the reference point for describing locations relative to the body's center.

| PLANES AND WHAT EACH ONE DIVIDES THE BODY INTO |                                   |                                                  |
|------------------------------------------------|-----------------------------------|--------------------------------------------------|
| Plane                                          | Orientation                       | Divides the body into                            |
| <b>Sagittal</b>                                | Vertical                          | Left and right parts                             |
| <b>Midsagittal (median)</b>                    | Vertical, on the midline          | Equal left and right halves                      |
| <b>Parasagittal</b>                            | Vertical, off the midline         | Unequal left and right parts                     |
| <b>Frontal (coronal)</b>                       | Vertical                          | Anterior and posterior portions                  |
| <b>Transverse (horizontal)</b>                 | Horizontal                        | Superior and inferior portions, a cross section  |
| <b>Oblique</b>                                 | At an angle other than 90 degrees | Angled portions, between horizontal and vertical |

## Directional Terms

Directional terms come in pairs of opposites, and they are always given from the patient's perspective in anatomical position. A structure lying between a medial and a lateral structure is called **intermediate**.

**PAIRED DIRECTIONAL TERMS**

| Term                      | Its opposite       | The relationship the pair describes                                                |
|---------------------------|--------------------|------------------------------------------------------------------------------------|
| <b>Superior</b>           | Inferior           | Toward the head or above, versus away from the head or below                       |
| <b>Anterior (ventral)</b> | Posterior (dorsal) | Toward the front, versus toward the back of the body                               |
| <b>Medial</b>             | Lateral            | Toward the midline, versus away from the midline                                   |
| <b>Proximal</b>           | Distal             | Closer to the point of attachment to the trunk, versus farther from it. Limbs only |
| <b>Superficial</b>        | Deep               | Toward the body surface, versus away from the surface                              |
| <b>Central</b>            | Peripheral         | Near the center of a structure, versus away from its center                        |
| <b>Ipsilateral</b>        | Contralateral      | On the same side of the body, versus on the opposite side                          |

**Regional Terms**

The body divides into an **axial** region, the head, neck, and trunk, and an **appendicular** region, the limbs. Each named region has a precise term. Learn these as a location on your own body, not as a word list.

**Anterior view****ANTERIOR REGIONAL TERMS**

| Term                     | Region                               | Term                         | Region                       |
|--------------------------|--------------------------------------|------------------------------|------------------------------|
| <b>Frontal</b>           | Forehead                             | <b>Inguinal</b>              | Groin                        |
| <b>Cranial</b>           | Skull or head                        | <b>Pubic</b>                 | Area near the genital region |
| <b>Facial</b>            | Face                                 | <b>Brachial</b>              | Upper arm                    |
| <b>Orbital or ocular</b> | Eye region                           | <b>Antebrachial</b>          | Forearm                      |
| <b>Buccal</b>            | Cheek                                | <b>Antecubital</b>           | Front of the elbow           |
| <b>Otic</b>              | Ear region                           | <b>Carpal</b>                | Wrist                        |
| <b>Nasal</b>             | Nose                                 | <b>Pollex</b>                | Thumb                        |
| <b>Oral</b>              | Mouth region                         | <b>Palmar</b>                | Palm of the hand             |
| <b>Mental</b>            | Chin                                 | <b>Digital or phalangeal</b> | Fingers or toes              |
| <b>Cervical</b>          | Neck region                          | <b>Patellar</b>              | Kneecap                      |
| <b>Thoracic</b>          | Chest area                           | <b>Crural</b>                | Leg, the shin area           |
| <b>Mammary</b>           | Breast region                        | <b>Tarsal</b>                | Ankle                        |
| <b>Abdominal</b>         | Below the chest and above the pelvis | <b>Pedal</b>                 | Foot                         |
| <b>Umbilical</b>         | Navel or belly button                | <b>Hallux</b>                | Big toe                      |
| <b>Pelvic</b>            | Lower abdomen or hip area            |                              |                              |

## Posterior view

| POSTERIOR REGIONAL TERMS |                         |           |                       |
|--------------------------|-------------------------|-----------|-----------------------|
| Term                     | Region                  | Term      | Region                |
| Cephalic                 | Head                    | Sacral    | Area between the hips |
| Cervical                 | Neck                    | Gluteal   | Buttock region        |
| Acromial                 | Point of the shoulder   | Femoral   | Thigh                 |
| Dorsal                   | Back                    | Popliteal | Back of the knee      |
| Olecranal                | Back of the elbow       | Sural     | Calf                  |
| Lumbar                   | Lower back or loin area | Calcaneal | Heel                  |
| Plantar                  | Sole of the foot        |           |                       |

## Clinical Vocabulary

### Disease versus disorder

**Disease** is a pathological condition caused by identifiable physical or biochemical changes, often from infection, genetics, or environmental factors. A disease typically has a clear set of symptoms, a predictable progression, and an identifiable cause. Examples: diabetes mellitus, a metabolic disease involving insulin production or utilization. Tuberculosis, an infectious disease caused by *Mycobacterium tuberculosis*.

**Disorder** is a disruption of normal physical or mental function that often lacks a single clear cause. Disorders may result from complex interactions among genetic, environmental, and lifestyle factors. Examples: anxiety disorder, marked by excessive fear or worry. Functional bowel disorders such as irritable bowel syndrome, which have no clear structural abnormality.

| TWO MORE WAYS TO SORT A CONDITION |                                                         |                          |
|-----------------------------------|---------------------------------------------------------|--------------------------|
| Term                              | Meaning                                                 | Example                  |
| Local                             | Affects one part or one limited region of the body      | A skin abscess           |
| Systemic                          | Affects the whole body or several organ systems at once | Sepsis, influenza        |
| Acute                             | Sudden onset, short duration                            | Appendicitis             |
| Chronic                           | Gradual onset, long duration, often lifelong            | Type 2 diabetes mellitus |

### Signs versus symptoms

A **sign** is objective. It is a measurable physical manifestation that a healthcare professional can observe. Fever measured as an elevated temperature, a visible rash, an increased heart rate on a pulse check or ECG.

A **symptom** is subjective. It is the patient's own experience of illness, and it cannot be measured directly by an observer. Pain, fatigue, nausea.



## THE SAME CONDITION, SORTED BOTH WAYS

| Condition        | Signs (observed)                               | Symptoms (reported)                        |
|------------------|------------------------------------------------|--------------------------------------------|
| Influenza        | Fever, nasal discharge, coughing               | Fatigue, body aches, sore throat           |
| Anxiety disorder | Increased heart rate, sweating during episodes | Feeling of dread, difficulty concentrating |

## What goes into a clinical diagnosis

- **History**, what the patient reports. Symptoms, timing, what makes it better or worse, past conditions, family history.
- **Physical examination**, the signs the clinician can observe, palpate, percuss, or listen for.
- **Vital signs**, temperature, pulse, respiratory rate, blood pressure, oxygen saturation.
- **Laboratory testing**, blood, urine, culture, tissue.
- **Imaging**, radiograph, CT, MRI, ultrasound. The plane you image in determines what you can see.
- **Differential diagnosis**, the ranked list of conditions that could explain the findings, narrowed as evidence comes in.

## WHY THE WORDS ARE NOT OPTIONAL

Anatomical terms remove ambiguity. A chart that reads "wound on the anterior, distal forearm" means the same thing to every clinician on every shift, no matter how the patient is positioned. Imaging is built on the planes: a CT slice is a transverse section, and an MRI is often read in the sagittal and frontal planes, so knowing the plane tells you which view you are looking at. Paired terms have to be used precisely. Proximal and distal describe the limbs only, never the trunk. Ipsilateral and contralateral matter in neurology, where a lesion on one side of the brain can show its effect on the opposite side of the body.

## SECTION 2

## Body Cavities &amp; Regions

## INTRODUCTION TO ANATOMY

The body cavities enclose and organize the internal organs. Knowing which cavity a structure sits in tells you what membrane wraps it, what fluid surrounds it, and what happens when that space is invaded.

**COMPETENCIES** W1-BODY-CAVITIES, W1-SEROUS-MEMBRANES, W1-MEDIASTINUM, W1-PERITONEAL-RELATIONSHIPS, W1-ABDOMINOPELVIC-REGIONS

## BY THE END

1. Name the two major body cavities and their subdivisions, and state what each contains.
2. Define a serous membrane and name the three membrane pairs, their spaces, and their fluids.
3. Give the boundaries and contents of the mediastinum.
4. Define retroperitoneal and list the retroperitoneal structures using SAD PUCKER.
5. Draw the four quadrants and the nine regions and name the organs in each.
6. Connect an inflamed serous membrane to the condition it produces.



WATCH AFTER YOU READ THIS CHAPTER

**Concept videos: Week 1**

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## The Two Major Body Cavities

A **body cavity** is a space within the body that houses and protects internal organs. There are two major cavities, and each divides further.

### CAVITIES, SUBDIVISIONS, AND CONTENTS

| Major cavity       | Subdivision               | Contents                        |
|--------------------|---------------------------|---------------------------------|
| Anterior (ventral) | Thoracic cavity           | The lungs and the heart         |
| Anterior (ventral) | Abdominopelvic cavity     | The abdominal and pelvic organs |
| Posterior (dorsal) | Cranial cavity            | The brain                       |
| Posterior (dorsal) | Spinal (vertebral) cavity | The spinal cord                 |

**The boundary.** The **diaphragm** is the muscular partition that separates the thoracic cavity above from the abdominopelvic cavity below. It is the single most useful landmark in the ventral cavity, because it tells you which membranes and which fluids you are dealing with.

## Serous Membranes

A **serous membrane** is a membrane that lines a cavity lacking an opening to the outside and covers the internal organs within it. Every serous membrane is a double layer folded back on itself.

- **Parietal layer**, lines the inner body wall.
- **Visceral layer**, covers the surface of an organ.
- **Serous space**, the cavity between the two layers.
- **Serous fluid**, found in that space between the visceral and parietal layers. It reduces friction so the organ can move against the wall without damage.

### THE THREE NAMED SEROUS MEMBRANE PAIRS

| Membrane    | Wraps                                          | Space              | Fluid             |
|-------------|------------------------------------------------|--------------------|-------------------|
| Pericardium | The heart (parietal and visceral pericardium)  | Pericardial cavity | Pericardial fluid |
| Pleura      | The lungs (parietal and visceral pleura)       | Pleural cavity     | Pleural fluid     |
| Peritoneum  | The abdomen (parietal and visceral peritoneum) | Peritoneal cavity  | Peritoneal fluid  |

### HOLD ONTO THIS

Parietal goes with the wall. Visceral goes with the organ. The two words never change meaning, no matter which cavity you are in.

## The Mediastinum

The **mediastinum** is the central compartment of the thoracic cavity, the region between the two lungs.

**BOUNDARIES OF THE MEDIASTINUM**

| Boundary  | Structure        |
|-----------|------------------|
| Superior  | Thoracic inlet   |
| Inferior  | Diaphragm        |
| Anterior  | Sternum          |
| Posterior | Vertebral column |
| Lateral   | Lungs            |

**Contents:** heart, trachea, esophagus, major blood vessels, thymus.

### The Peritoneum and Retroperitoneal Structures

The **retroperitoneum** is the area posterior to the peritoneum but anterior to the spine. A structure that lies there is called **retroperitoneal**. A structure fully wrapped in visceral peritoneum and suspended within the peritoneal cavity is **intra-peritoneal**. The distinction is surgical: it determines the approach a surgeon takes and how an infection or a bleed will spread.

**MNEMONIC: SAD PUCKER**

**S** Suprarenal (adrenal) glands. **A** Aorta and inferior vena cava. **D** Duodenum, second and third segments. **P** Pancreas, except the tail. **U** Ureters. **C** Colon, ascending and descending. **K** Kidneys. **E** Esophagus. **R** Rectum.

### Abdominal Quadrants and Regions

#### The four quadrants

Two planes draw the quadrant grid: a median (midsagittal) plane and a transverse plane through the umbilicus. Quadrants are the fast clinical shorthand, the version used when someone is describing pain in a triage note.

**THE FOUR QUADRANTS AND THEIR KEY ORGANS**

| Quadrant          | Key organs         |
|-------------------|--------------------|
| Right upper (RUQ) | Liver, gallbladder |
| Left upper (LUQ)  | Stomach, spleen    |
| Right lower (RLQ) | Appendix, cecum    |
| Left lower (LLQ)  | Descending colon   |

#### The nine regions

Four planes, two vertical and two horizontal, divide the abdominopelvic cavity into nine regions. Regions are the more precise anatomical version, the one used when the location has to be described exactly.

**THE NINE ABDOMINOPELVIC REGIONS**

| Region              | Key structure      |
|---------------------|--------------------|
| Right hypochondriac | Liver, gallbladder |
| Epigastric          | Stomach            |
| Left hypochondriac  | Spleen             |
| Right lumbar        | Ascending colon    |
| Umbilical           | Small intestine    |
| Left lumbar         | Descending colon   |
| Right iliac         | Cecum, appendix    |
| Hypogastric         | Urinary bladder    |
| Left iliac          | Sigmoid colon      |

**Inflammatory Disorders of the Serous Membranes**

Each named membrane has a matching inflammatory condition. If you know the membrane and its location, you can predict the presentation.

**FOUR CONDITIONS, BY MEMBRANE**

| Condition           | Causes                                | Signs and symptoms                   | Treatment                                            |
|---------------------|---------------------------------------|--------------------------------------|------------------------------------------------------|
| <b>Peritonitis</b>  | Infection, trauma, ruptured appendix  | Abdominal pain, fever, rigidity      | Antibiotics, surgery                                 |
| <b>Pleurisy</b>     | Viral infection, pneumonia            | Sharp chest pain, dry cough          | Anti-inflammatory drugs                              |
| <b>Pericarditis</b> | Viral infection, autoimmune disorders | Chest pain, pericardial friction rub | NSAIDs, corticosteroids                              |
| <b>Meningitis</b>   | Bacterial or viral infection          | Stiff neck, fever, headache          | Antibiotics for bacterial, supportive care for viral |

**THE CAVITY TELLS YOU THE CONSEQUENCES**

A serous cavity is a closed space, so anything that enters it has nowhere to drain. A ruptured appendix spills bowel contents into the peritoneal cavity, and the result is peritonitis across the whole abdomen rather than a local problem. A puncture of the pleural cavity lets air in, the negative pressure that holds the lung expanded is lost, and the lung collapses. Reduced serous fluid in any of the three spaces means the two layers rub instead of glide, which is why a friction rub is audible in pericarditis. Meningitis is the same logic applied to the dorsal cavity: the meninges enclose the brain and spinal cord, so an infection there raises pressure inside a rigid skull.

**SECTION 3****Anatomy of the Cell****CELL ANATOMY**

The cell is the smallest living unit of the body. Every organelle in it is a structure with a job, and every one of those jobs has a disease that shows what happens when it fails.

COMPETENCIES W1-GENERALIZED-CELL, W1-PLASMA-MEMBRANE, W1-NUCLEUS, W1-ORGANELLE-ID

## BY THE END

1. Identify the three major components of a cell and describe what each does.
2. Define semipermeable, permeable, and impermeable, with examples.
3. Name the structures of the nucleus and state the function of each.
4. Separate cytoplasm from cytosol.
5. Describe the plasma membrane and the fluid mosaic model.
6. List the components of the endomembrane system and trace a protein through it.
7. Describe each organelle and connect it to a clinical consequence when it fails.



WATCH AFTER YOU READ THIS CHAPTER

**Concept videos: The Cell**
[drsrennie-stack.github.io/new-build-bio4-solano/cell-concept-videos.html](https://drsrennie-stack.github.io/new-build-bio4-solano/cell-concept-videos.html)

## General Structure of the Cell

### Plasma membrane

A flexible, semipermeable boundary that encloses the cell, separates its contents from the external environment, and regulates the passage of substances. It is composed primarily of lipids and proteins with some carbohydrates, and it carries specialized structures such as receptors, ion channels, and transport proteins for communication and material exchange.

### Nucleus

The largest organelle and the command center of the cell. It contains the genetic material, DNA, organized into chromosomes. It controls gene expression, cellular growth, and replication, and it is the site of transcription, the conversion of DNA to RNA.

### Cytoplasm

The gel-like substance within the plasma membrane, excluding the nucleus, containing the organelles and the cytosol. It facilitates intracellular processes such as glycolysis and signal transduction, helps maintain cell shape, and enables the movement of materials.

## Cell Permeability

### THREE DEGREES OF PERMEABILITY

| Term                 | Definition                                                                                                                                                               | Examples                                                                 |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>Semipermeable</b> | Allows selective substances to cross based on size, charge, and solubility. Facilitates passive diffusion for some molecules and relies on transport proteins for others | Oxygen, carbon dioxide, and small lipid-soluble molecules diffuse freely |
| <b>Permeable</b>     | Permits the free passage of substances without restriction                                                                                                               | Water through aquaporins, gases such as nitrogen                         |
| <b>Impermeable</b>   | Blocks movement unless a specific mechanism carries the substance across                                                                                                 | Proteins, polysaccharides, charged ions with no transporter              |

## The Nucleus in Detail

**Multinucleated cells** have more than one nucleus. Skeletal muscle fibers and hepatocytes in the liver are the standard examples. **Anucleated cells** have none. Mature red blood cells are the example, and losing the nucleus is what makes room for hemoglobin.

**Function.** The nucleus stores the DNA that carries instructions for protein synthesis and cellular activities, plays a central role in cell division, growth, and metabolism, and regulates which genes are turned on or off in different cells. That last one is what allows differentiation.

#### STRUCTURES OF THE NUCLEUS

| Structure               | What it is and does                                                                                          |
|-------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Nuclear envelope</b> | A double-layered membrane separating the nucleus from the cytoplasm and maintaining a controlled environment |
| <b>Nuclear pores</b>    | Protein-lined channels allowing exchange of materials such as RNA and proteins between nucleus and cytoplasm |
| <b>Chromatin</b>        | A network of DNA and histone proteins that condenses into visible chromosomes during cell division           |
| <b>Chromosomes</b>      | Rod-shaped structures containing genes, responsible for passing genetic information during reproduction      |
| <b>Nucleoplasm</b>      | The viscous fluid within the nucleus supporting the chromatin and nucleolus                                  |
| <b>Nucleolus</b>        | A dense region where ribosomal RNA is synthesized and ribosomal subunits are assembled                       |

### Cytoplasm versus Cytosol

**Cytoplasm** is the entire contents of the cell excluding the nucleus, meaning the organelles suspended in cytosol.

**Cytosol** is the aqueous component alone: water, ions, enzymes, and small organic molecules. The cytosol is where many metabolic reactions happen, glycolysis among them.

### The Plasma Membrane in Detail

**Function.** A dynamic barrier that regulates interactions between the cell and its environment.

#### MEMBRANE COMPONENTS

| Component                            | Arrangement and role                                                                                                                   |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phospholipid bilayer</b>          | Hydrophilic heads face outward toward the extracellular and intracellular fluids. Hydrophobic tails face inward, away from water       |
| <b>Proteins</b>                      | Integral proteins span the bilayer, peripheral proteins bind to the surface. Roles: transport, enzymatic activity, signal transduction |
| <b>Glycoproteins and glycolipids</b> | Cell recognition and immune responses                                                                                                  |
| <b>Cholesterol</b>                   | Provides stability and moderates membrane fluidity at varying temperatures                                                             |

**Fluid mosaic model.** The membrane behaves as a fluid-like sea of phospholipids with proteins and carbohydrates floating within it. Proteins and lipids move laterally, which allows flexibility and dynamic interaction rather than a fixed wall.

### The Endomembrane System

A coordinated network of organelles responsible for protein and lipid synthesis, modification, and transport.

- **Nuclear envelope**, protects the genetic material.
- **Rough ER**, synthesizes membrane-bound and secretory proteins.
- **Smooth ER**, synthesizes lipids, detoxifies chemicals, stores calcium.

- **Golgi apparatus**, modifies, sorts, and packages proteins and lipids.
- **Lysosomes**, digestive organelles containing hydrolytic enzymes.
- **Vesicles**, transport materials between organelles or to the plasma membrane.

## Ribosomes

**Function.** The site of protein synthesis, where mRNA is translated into polypeptides. **Structure.** A large and a small subunit, each made of ribosomal RNA and protein.

- **Free ribosomes**, produce proteins for use inside the cell.
- **Bound ribosomes**, attached to the rough ER, producing proteins for secretion or for insertion into membrane.
- **Mitochondrial ribosomes**, produce proteins specific to mitochondrial function.

## Endoplasmic Reticulum

### ROUGH AND SMOOTH ER COMPARED

|                  | Rough ER                                          | Smooth ER                                                                            |
|------------------|---------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Structure</b> | Studded with ribosomes                            | Lacks ribosomes                                                                      |
| <b>Function</b>  | Produces membrane proteins and secretory proteins | Synthesizes lipids, metabolizes carbohydrates, detoxifies drugs, stores calcium ions |
| <b>Next stop</b> | Sends vesicles to the Golgi apparatus             | Supplies lipid and calcium to the rest of the cell                                   |

## Golgi Apparatus

**Function.** Modifies, packages, and sorts proteins and lipids, tagging molecules for delivery to specific cellular locations. **Structure.** A series of flattened sacs called cisternae, with vesicles budding off.

## Lysosomes

**Function.** Digestive organelles that break down macromolecules, damaged organelles, and pathogens. Two processes to know: **phagocytosis**, engulfing pathogens or debris, and **autophagy**, recycling of the cell's own components.

### CLINICAL CORRELATION

In Tay-Sachs disease, a deficiency of a lysosomal enzyme causes toxic accumulation of lipids in nerve cells. The organelle that should be clearing material cannot, so the material builds up inside the cell until the cell fails.

## Peroxisomes

**Function.** Break down fatty acids, amino acids, and hydrogen peroxide. They also detoxify alcohol in the liver and kidney, which is why those organs are peroxisome-rich.

## Mitochondria

**Function.** The powerhouse of the cell, producing ATP through cellular respiration. **Structure.** A double membrane with inner folds called cristae, which multiply the surface area available for ATP production. Mitochondria contain their own DNA and ribosomes, so they produce some of their own proteins.

## Cytoskeleton

**Function.** Provides structural support, facilitates intracellular transport, and enables cellular movement.

**THREE CYTOSKELETAL COMPONENTS**

| Component              | Made of                  | Role                                                       |
|------------------------|--------------------------|------------------------------------------------------------|
| Microfilaments         | Actin                    | Cell shape and movement                                    |
| Intermediate filaments | Various fibrous proteins | Mechanical support, resisting tension                      |
| Microtubules           | Tubulin                  | Tubular structures forming centrioles, cilia, and flagella |

**Centrioles and Centrosomes**

**Centrioles** are short cylindrical structures organized in a 9 plus 0 microtubule arrangement. The **centrosome** is the microtubule-organizing center. Both are essential for forming the mitotic spindle during cell division.

**Cilia and Flagella**

**Cilia** are short projections that move substances across the cell surface, such as mucus in the respiratory tract. **Flagella** are long projections that propel the cell itself, as in sperm. Both use a 9 plus 2 microtubule arrangement, and that arrangement is what makes movement possible.

**HOLD ONTO THIS**

9 plus 0 does not move, it organizes. 9 plus 2 moves. Centrioles are 9 plus 0. Cilia and flagella are 9 plus 2.

**THE ORGANELLE POINTS TO THE DISEASE**

Read this list backward when you are given a case. A cell that cannot break down lipids has a lysosome problem. A cell that cannot make enough ATP has a mitochondrial problem, and the tissues that fail first are the ones with the highest energy demand: muscle, brain, kidney. A cell that cannot clear mucus has a ciliary problem, and the presentation is recurrent respiratory infection. Structure first, then the consequence.

**SECTION 4****Histology: The Four Tissue Types****TISSUES & HISTOLOGY**

Tissues are groups of similar cells performing a specific function. Four types build every organ in the body, and the arrangement of the cells is what predicts what the tissue can do.

**COMPETENCIES** W1-GERM-LAYERS, W1-EPITHELIAL-ID, W1-CELL-JUNCTIONS, W1-CONNECTIVE-ID, W1-BODY-MEMBRANES

**BY THE END**

1. Name the four tissue types, their general functions, and where each is found.
2. Define a biopsy and walk through the procedure.
3. Identify the five cell junctions, their proteins, locations, and functions.
4. Separate epithelial from connective tissue on structure alone.
5. Classify epithelium by layers and cell shape, and name a location for each type.
6. Classify glands structurally and functionally.
7. Name the cells, fibers, and ground substance of connective tissue and what each contributes.
8. Recognize cartilage, bone, muscle, and nervous tissue at survey level.





WATCH AFTER YOU READ THIS CHAPTER

**Concept videos: Tissues**

[drsrennie-stack.github.io/new-build-bio4-solano/tissues-concept-videos.html](https://drsrennie-stack.github.io/new-build-bio4-solano/tissues-concept-videos.html)

## The Four Major Tissue Types

1. **Epithelial tissue**, covers body surfaces, lines cavities, and forms glands.
2. **Connective tissue**, protects, supports, and binds structures and organs.
3. **Nervous tissue**, detects changes and generates nerve impulses.
4. **Muscle tissue**, facilitates movement and generates force.

### Cells that stay and cells that move

Most cells are **stationary**. Some **migrate**, particularly during growth and development. White blood cells migrating to a site of injury or infection are the everyday example.

## Biopsy

**Definition.** Removal of a tissue sample for microscopic examination. This is how tissue-level questions get answered clinically.

1. Collect a tissue sample using a needle, scalpel, or endoscope.
2. Preserve the tissue in a fixative such as formalin.
3. Section the tissue into thin slices for examination.
4. Stain the tissue for contrast under a microscope.

## The Five Cell Junctions

| JUNCTION, PROTEIN, LOCATION, FUNCTION |                                                                                                                           |                                                                                                               |                                                                                 |                        |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------|
| Junction                              | What it is                                                                                                                | Function                                                                                                      | Location                                                                        | Major protein          |
| <b>Tight junction</b>                 | A network of transmembrane proteins that fuse together on adjacent plasma membranes, sealing the passageway between cells | Prevents substances passing between cells, prevents organ contents leaking into surrounding tissue            | Epithelium lining the stomach, intestines, urinary bladder, blood-brain barrier | Claudins and occludins |
| <b>Adherens junction</b>              | Adhesion belts formed by transmembrane proteins attached to microfilaments                                                | Provides mechanical stability, helps epithelial surfaces resist separation during contractile activity        | Intestinal epithelial cells                                                     | Cadherins              |
| <b>Desmosome</b>                      | Plaques and cadherins connected to intermediate filaments                                                                 | Strongly binds cells to each other, prevents separation under tension                                         | Outer epidermis, cardiac muscle cells                                           | Cadherins              |
| <b>Hemidesmosome</b>                  | Anchors cells to the basement membrane                                                                                    | Prevents separation of epidermis from basement membrane, stabilizes epithelial cells during mechanical stress | Skin, heart muscle                                                              | Integrins              |
| <b>Gap junction</b>                   | Protein channels connecting neighboring cells, allowing passage of small ions and molecules                               | Cell to cell communication, transfers nutrients and waste, allows impulses to spread                          | Lens, cornea, embryonic tissues, neurons, cardiac tissue, GI tract, uterus      | Connexins              |

### HOLD ONTO THIS

Desmosomes connect cell to cell. Hemidesmosomes connect cell to basement membrane. Half the name, half the job.

## Epithelium versus Connective Tissue

| THE TWO TISSUE TYPES COMPARED |                                                    |                                                          |
|-------------------------------|----------------------------------------------------|----------------------------------------------------------|
| Feature                       | Epithelial tissue                                  | Connective tissue                                        |
| <b>Matrix and cells</b>       | Tightly packed cells, minimal extracellular matrix | Large extracellular matrix, few cells                    |
| <b>Blood vessels</b>          | Avascular, relies on diffusion                     | Highly vascular, with cartilage and tendon as exceptions |
| <b>Location</b>               | Covers surfaces and forms linings                  | Supports and connects other tissues                      |

**Similarities.** They work together for nutrient exchange, since avascular epithelium depends on the vascular connective tissue beneath it. Both play critical roles in maintaining structure and function.

## Epithelial Tissue

### Three functions of epithelium

- **Selective barrier**, controls transfer of substances into and out of the body.
- **Secretory surface**, produces and releases cellular products.
- **Protective surface**, provides resistance against abrasion and environmental influences.

### The basement membrane

**Location.** Below the basal surface of epithelial tissue. **Composition.** A **basal lamina** of collagen, laminin, and glycoproteins, plus a **reticular lamina** of collagen and glycoproteins.

**Functions.** Supports and anchors epithelial tissue, facilitates wound healing and cell migration, restricts large molecules, and filters substances in the kidneys.

### Classification of epithelium

| EPITHELIAL TYPES, FUNCTION, AND LOCATION |                                                                                               |                                                  |                                                     |
|------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------|
| Tissue type                              | Description                                                                                   | Function                                         | Locations                                           |
| Simple squamous                          | Single layer of flat, thin cells                                                              | Filtration, diffusion, osmosis, secretion        | Alveoli, glomeruli, lining of blood vessels, serosa |
| Simple cuboidal                          | Single layer of cube-shaped cells                                                             | Secretion and absorption                         | Kidney tubules, small glands, ovary surface         |
| Simple columnar                          | Single layer of tall, column-shaped cells, may have microvilli or goblet cells                | Absorption, secretion of mucus and enzymes       | Digestive tract from stomach to rectum, gallbladder |
| Stratified squamous                      | Several layers, apical layers are flat                                                        | Protection against abrasion                      | Skin (keratinized), esophagus, mouth, vagina        |
| Stratified cuboidal                      | Two or more layers of cube-shaped cells                                                       | Protection and limited secretion                 | Sweat glands, mammary glands, salivary glands       |
| Stratified columnar                      | Several layers, apical cells are columnar                                                     | Protection and secretion                         | Male urethra, ducts of some glands                  |
| Transitional                             | Several layers, cells stretch and change shape from dome-shaped to flat                       | Stretching to allow distension                   | Urinary bladder, ureters, part of the urethra       |
| Pseudostratified ciliated columnar       | Appears stratified but all cells touch the basement membrane, may have cilia and goblet cells | Secretion of mucus, propulsion by ciliary action | Respiratory tract (trachea, bronchi), sperm ducts   |

#### HOLD ONTO THIS

Name the layers first, then the shape of the apical cells. Simple means one layer, stratified means more than one, and the shape you name is always the shape at the free surface.

## Glandular Epithelium

### Structural classification

#### GLANDS BY NUMBER OF CELLS

| Type          | Description                        | Example                           |
|---------------|------------------------------------|-----------------------------------|
| Unicellular   | Single-celled glands               | Goblet cells, which secrete mucus |
| Multicellular | Multiple cells forming a structure | Salivary glands, sweat glands     |

**Duct type.** Simple means no branching. Compound means branching. **Shape of the secretory portion.** Tubular means tube-shaped. Acinar means rounded. Combine the two and you have the full structural name.

### Endocrine and exocrine

**Endocrine glands** release their product into the bloodstream, with no duct. Thyroid, adrenal, pituitary. **Exocrine glands** secrete their products into ducts that empty onto a surface or into a lumen. Sweat, salivary, sebaceous. Some glands are **mixed**, doing both, and the pancreas is the classic example: exocrine digestive enzymes into a duct, endocrine insulin and glucagon into the blood.

### Functional classification of exocrine glands

#### HOW THE PRODUCT ACTUALLY LEAVES THE CELL

| Type      | Mechanism                                                                                                           | Example                   |
|-----------|---------------------------------------------------------------------------------------------------------------------|---------------------------|
| Merocrine | Secretions released via exocytosis. The cell is unharmed. Most common type                                          | Salivary glands, pancreas |
| Apocrine  | Secretions accumulate at the apical surface, which pinches off to release the product. The cell then repairs itself | Mammary glands            |
| Holocrine | Secretions accumulate in the cytoplasm and the entire cell ruptures to release the product                          | Sebaceous glands          |

#### MEMORY TOOL

HOL equals WHOLE cell is secreted.

## Connective Tissue

### Functions

- Binds together, supports, and strengthens body tissues.
- Protects and insulates internal organs.
- Compartmentalizes structures, as in skeletal muscle.
- Serves as a major transport system. Blood is a connective tissue.
- Is the primary location of stored energy reserves, as in adipose tissue.
- Is a main source of immune responses.

### The two basic elements

Connective tissue is **extracellular matrix** plus **cells**. The matrix divides into **ground substance** and **protein fibers**. The proportions of these elements are what make blood liquid, cartilage firm, and bone hard.

## Ground substance

**Definition.** The material between cells and protein fibers. **Structure.** Fluid, semifluid, gelatinous, or calcified.

**Composition.** Water and organic molecules such as chondroitin sulfate and glucosamine.

**Functions.** Supports and binds cells, supplies water and nutrients for exchange between blood and cells, and plays a role in tissue development and metabolism.

**The matrix determines the quality.** Cartilage is firm but pliable. Bone is hard and inflexible. Same basic plan, different matrix.

## The three fiber types

| COLLAGEN, RETICULAR, AND ELASTIC FIBERS |                                                                                   |                                                                             |                                                                                                                                                                                                    |
|-----------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fiber                                   | Protein                                                                           | Properties                                                                  | Locations and functions                                                                                                                                                                            |
| <b>Collagen</b>                         | Collagen, the most abundant protein in the body at about 25 percent               | Very strong, resists tension. Parallel bundles provide tensile strength     | Most connective tissues: bone, tendon, cartilage                                                                                                                                                   |
| <b>Reticular</b>                        | Collagen covered with glycoproteins                                               | Fine, branching networks                                                    | Walls of blood vessels, surrounding cells in areolar, adipose, nervous, and smooth muscle tissue. Provides support and strength, forms the stroma of soft organs, helps form the basement membrane |
| <b>Elastic</b>                          | Elastin, with the glycoprotein fibrillin providing tensile strength and stability | Stretches up to 150 percent of relaxed length and returns to original shape | Skin, lungs, blood vessel walls                                                                                                                                                                    |

## Cells of connective tissue

| WHO LIVES IN THE MATRIX  |                                                       |                                                                                                                                                                                     |
|--------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cell                     | Origin or definition                                  | Function and location                                                                                                                                                               |
| <b>Fibroblasts</b>       | Found in all general connective tissues               | Migrate through connective tissue secreting matrix fibers and ground substance                                                                                                      |
| <b>Macrophages</b>       | Irregular white blood cells derived from monocytes    | Engulf bacteria and debris via phagocytosis. Fixed macrophages stay in one tissue such as liver, lungs, spleen. Wandering macrophages migrate to sites of infection or inflammation |
| <b>Mast cells</b>        | Found along blood vessels supplying connective tissue | Secrete histamine, which dilates blood vessels during inflammation. Also bind, ingest, and kill bacteria                                                                            |
| <b>Plasma cells</b>      | Develop from B lymphocytes                            | Secrete antibodies. Found in connective tissue of the GI tract, respiratory tract, and lymphatic tissues                                                                            |
| <b>Adipocytes</b>        | Connective tissue cells that store triglycerides      | Found deep in the skin and around organs such as the heart and kidneys                                                                                                              |
| <b>White blood cells</b> | Various leukocytes                                    | Immune defense                                                                                                                                                                      |

## Blasts versus cytes

### IMMATURE AND MATURE FORMS

|                     | Immature, the blasts                             | Mature, the cytes                        |
|---------------------|--------------------------------------------------|------------------------------------------|
| <b>Examples</b>     | Fibroblasts, osteoblasts, chondroblasts          | Fibrocytes, osteocytes, chondrocytes     |
| <b>What they do</b> | Actively divide and secrete extracellular matrix | Maintain and repair extracellular matrix |

## General features of connective tissue

- **Location.** Found below epithelial tissue, never on body surfaces.
- **Vascularity.** Highly vascular. Exceptions: cartilage and tendon.
- **Innervation.** Supplied with nerves. Exceptions: cartilage and blood.

## Embryonic mesenchyme and mucous connective tissue

### THE TWO EMBRYONIC CONNECTIVE TISSUES

|                        | Embryonic mesenchyme                                                 | Mucous connective tissue                            |
|------------------------|----------------------------------------------------------------------|-----------------------------------------------------|
| <b>Location</b>        | In the embryo, under skin and along bones                            | In the umbilical cord                               |
| <b>Significance</b>    | Differentiates into all types of connective tissue                   | Provides support                                    |
| <b>Characteristics</b> | Irregular cells in a semifluid matrix with delicate reticular fibers | Widely scattered fibroblasts in a jelly-like matrix |

## Blood, a Fluid Connective Tissue

### FORMED ELEMENTS AND WHAT EACH DOES

| Cell type                        | Function                                  |
|----------------------------------|-------------------------------------------|
| <b>Neutrophils</b>               | Phagocytosis of bacteria                  |
| <b>Eosinophils</b>               | Combat parasitic infections and allergies |
| <b>Lymphocytes</b>               | Fight viruses                             |
| <b>Monocytes and macrophages</b> | Phagocytosis, transform into macrophages  |
| <b>Basophils</b>                 | Involved in allergic responses            |
| <b>Red blood cells</b>           | Transport oxygen and carbon dioxide       |
| <b>Platelets</b>                 | Facilitate blood clotting                 |

## Connective Tissues, Cartilage, Bone, and Nervous Tissue

### SURVEY OF THE REMAINING TISSUE TYPES

| Tissue type               | Description                                                                                         | Function                                                   | Locations                                                   |
|---------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------|
| <b>Areolar connective</b> | Loose arrangement of collagen, elastic, and reticular fibers with scattered cells                   | Strength, elasticity, support                              | Under epithelial tissue, around organs and blood vessels    |
| <b>Adipose connective</b> | Cells filled with fat droplets, sparse extracellular matrix                                         | Energy storage, insulation, cushioning                     | Subcutaneous tissue, around kidneys and heart               |
| <b>Hyaline cartilage</b>  | Glassy appearance, fine collagen fibers not visible with ordinary staining, chondrocytes in lacunae | Smooth surface for joint movement, flexible support        | Ends of long bones, costal cartilage, nose, trachea, larynx |
| <b>Fibrocartilage</b>     | Thick collagen fibers dominate                                                                      | Resists compression, absorbs shock                         | Intervertebral discs, pubic symphysis, knee menisci         |
| <b>Elastic cartilage</b>  | Similar to hyaline but with more elastic fibers                                                     | Maintains shape and provides flexibility                   | External ear (pinna), epiglottis                            |
| <b>Compact bone</b>       | Hard, calcified matrix with collagen fibers, osteocytes in lacunae                                  | Supports, protects, stores calcium, houses blood formation | Bones                                                       |
| <b>Nervous tissue</b>     | Neurons and glial cells, branching neurons with long processes                                      | Transmission of electrical signals                         | Brain, spinal cord, nerves                                  |

## Muscle Tissue

### THE THREE MUSCLE TISSUE TYPES

| Tissue type            | Description                                                             | Function                                 | Locations                                             |
|------------------------|-------------------------------------------------------------------------|------------------------------------------|-------------------------------------------------------|
| <b>Skeletal muscle</b> | Long, cylindrical, multinucleated cells, striated                       | Voluntary movement, heat production      | Attached to bones and occasionally skin               |
| <b>Cardiac muscle</b>  | Branching cells connected by intercalated discs, striated, uninucleated | Involuntary contraction to pump blood    | Heart walls                                           |
| <b>Smooth muscle</b>   | Spindle-shaped cells, non-striated, uninucleated                        | Involuntary movement, propels substances | Walls of hollow organs such as intestines and bladder |

### TISSUE STRUCTURE PREDICTS HOW IT HEALS AND HOW IT FAILS

Vascularity sets healing speed. Cartilage and tendon are avascular or nearly so, which is why a torn meniscus or a ruptured Achilles heals slowly while a skin wound closes in days. Junction type sets what tears. Loss of desmosomal adhesion in the epidermis produces blistering, because the cells separate from each other. Loss of hemidesmosomes separates the whole epidermis from the basement membrane, which is a deeper and more serious blister. Cell arrangement sets the barrier. Stratified squamous epithelium survives abrasion because it sheds its top layer. Simple squamous cannot, which is why the alveoli, one cell thick for gas exchange, are so easily damaged.

## SECTION 5

# The Integumentary System

## INTEGUMENTARY & SKIN

The skin is the largest organ in the body and the first place all four tissue types show up working together. It is also the organ you can read from across the room, which makes it the best teacher of the structure to consequence habit.

**COMPETENCIES** W1-SKIN-LAYERS, W1-EPIDERMAL-STRATA, W1-DERMIS-LAYERS, W1-SKIN-ACCESSORY

### BY THE END

1. Name the components and the six functions of the integumentary system.
2. List the strata of the epidermis deep to superficial and say which is missing in thin skin.
3. Name the four principal cells of the epidermis and what each does.
4. Describe melanin transfer and why it protects the nucleus.
5. Separate the papillary from the reticular region of the dermis.
6. Describe hair anatomy, the follicle, and the bulb.
7. Compare the four skin glands on distribution, secretion, and onset.
8. Name the components of a nail and their functions.
9. Read skin color as a diagnostic clue.



**WATCH AFTER YOU READ THIS CHAPTER**

**Concept videos: Integumentary**

[drsrennie-stack.github.io/new-build-bio4-solano/integumentary-concept-videos.html](https://drsrennie-stack.github.io/new-build-bio4-solano/integumentary-concept-videos.html)

## Components and Functions

**Components.** The skin, also called the cutaneous membrane, plus associated structures: hair, oil glands, sweat glands, sensory receptors, and nails.

**Dermatology** is the medical specialty dealing with the structure, function, and disorders of the integumentary system.

### THE SIX FUNCTIONS OF THE SKIN

| Function                        | How the structure delivers it                                              |
|---------------------------------|----------------------------------------------------------------------------|
| <b>Thermoregulation</b>         | Sweat glands and dermal blood vessels adjust heat loss                     |
| <b>Blood reservoir</b>          | Extensive dermal vascular network holds a substantial volume of blood      |
| <b>Protection</b>               | Keratin, lamellar granules, melanin, and immune cells                      |
| <b>Cutaneous sensations</b>     | Tactile discs, corpuscles of touch, hair root plexuses, free nerve endings |
| <b>Excretion and absorption</b> | Small amounts of waste out, lipid-soluble substances in                    |
| <b>Synthesis of vitamin D</b>   | UV exposure begins the synthesis pathway in the skin                       |

**General information.** The skin is the largest organ by weight, covering the external surface. In an adult the area is roughly 22 square feet, and it weighs 10 to 11 pounds, about 7 percent of total body weight.

## The Three Layers

1. **Epidermis**, part of the skin.
2. **Dermis**, part of the skin.
3. **Subcutaneous tissue**, not part of the skin.



## The Epidermis

The thinner, superficial portion, composed of epithelial tissue and avascular. Its composition is **keratinized stratified squamous epithelium**. Avascular matters: the epidermis depends entirely on diffusion from the dermis beneath it, which is why a deep burn that destroys dermal vessels cannot support regeneration.

### The four principal cells

| CELLS OF THE EPIDERMIS                               |                                                               |                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cell                                                 | Share and origin                                              | Function                                                                                                                                                                                                                                                                        |
| <b>Keratinocytes</b>                                 | 90 percent, arranged in 4 to 5 layers                         | Produce keratin, a tough fibrous protein protecting skin and underlying tissue from abrasion, heat, microbes, and chemicals. Also produce lamellar granules, which release a water-repellent sealant that decreases water entry and loss and inhibits entry of foreign material |
| <b>Melanocytes</b>                                   | 8 percent, develop from the ectoderm of the embryo            | Produce melanin, a yellow-red or brown-black pigment contributing to skin color and absorbing damaging UV light                                                                                                                                                                 |
| <b>Intraepidermal macrophages (Langerhans cells)</b> | Originate in red bone marrow                                  | Migrate to the epidermis and participate in immune responses against microbes invading the skin. Help other immune cells recognize and destroy an invader. Easily damaged by UV light                                                                                           |
| <b>Merkel cells (tactile cells)</b>                  | Least numerous, located in the deepest layer of the epidermis | Contact the flattened process of a sensory neuron, called a Merkel disc or tactile disc                                                                                                                                                                                         |

### The strata

Most regions of the body have **four strata**: stratum basale, stratum spinosum, stratum granulosum, stratum corneum. Areas of greatest friction, meaning fingertips, palms, and soles, have **five strata**, adding the stratum lucidum between the granulosum and the corneum.

#### MNEMONIC

Baby Screams Grandpa Loves Coffee. Basale, Spinosum, Granulosum, Lucidum, Corneum. Deep to superficial.

### Melanin transfer

1. Melanocytes have long slender projections that extend between keratinocytes and transfer melanin granules to them.
2. Once inside the keratinocyte, melanin granules cluster to form a protective veil over the nucleus, on the side facing the skin surface.
3. That veil shields the DNA, preventing UV damage from the sun.
4. Melanocytes themselves remain susceptible to damage by UV light. They protect everyone else and not themselves.

## The Dermis

The second, deeper portion of the skin. **Tissue composition.** Dense irregular connective tissue with collagen and elastic fibers.

**Functions.** Gives the skin great tensile strength, and allows stretch and recoil.

**Characteristics.** Thicker than the epidermis and variable by body site, greatest at the soles and palms. Contains blood vessels, nerves, glands, and hair follicles. **Cells.** Fibroblasts, a few macrophages, and a few adipocytes near the boundary with the subcutaneous layer. The dermis is essential to the survival of the epidermis and its associated structures.

| THE TWO PRINCIPAL LAYERS OF THE DERMIS |                                                                                                                                |                                                                                                                                                                                   |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | Papillary region                                                                                                               | Reticular region                                                                                                                                                                  |
| <b>Position and size</b>               | Thinner, superficial. One fifth of the total dermal layer                                                                      | Thicker, deep. Four fifths of the total dermal layer                                                                                                                              |
| <b>Contents</b>                        | Thin collagen and fine elastic fibers. Dermal papillae, small finger-shaped projections into the undersurface of the epidermis | Bundles of thick collagen fibers, scattered fibroblasts, wandering macrophages, adipocytes in the deepest part with coarse elastic fibers                                         |
| <b>Structures within</b>               | Capillary loops, Meissner corpuscles for touch and pressure, free nerve endings for temperature, light touch, and itching      | Blood vessels, nerves, hair follicles, sudoriferous sweat glands, sebaceous glands                                                                                                |
| <b>Function</b>                        | Anchors epidermis, supplies it by diffusion, carries the fine touch receptors                                                  | Net-like layer attached to the subcutaneous layer. Provides strength, extensibility, and elasticity. The regular net-like collagen orientation makes skin resistant to stretching |

### Epidermal ridges, the fingerprints

The bond between dermis and epidermis on the surfaces of the palms, soles, and fingertips. **Functions.** Increase surface area, which increases the number of corpuscles of touch and therefore tactile sensitivity, and increase grip by increasing friction.

### The Subcutaneous Layer (Hypodermis)

Deep to the dermis and not part of the skin. **Composition.** Areolar and adipose tissue. Fibers extend from the dermis and anchor the skin to this layer and to the fascia around muscles and bones. **Function.** Storage for fat. **Associated components.** Lamellated corpuscles, which are nerve endings sensitive to pressure, and large blood vessels.

### Skin Color

| THREE PIGMENTS    |                              |                                                                                              |
|-------------------|------------------------------|----------------------------------------------------------------------------------------------|
| Pigment           | Color contributed            | Notes                                                                                        |
| <b>Melanin</b>    | Pale yellow to reddish-brown | Two forms: pheomelanin, yellow to red, and eumelanin, brown to black                         |
| <b>Hemoglobin</b> | Pink to red                  | Seen through the epidermis, depends on the oxygen content of blood in the dermal capillaries |
| <b>Carotene</b>   | Yellow-orange                | A precursor to vitamin A, used to synthesize pigments for vision                             |

**Melanocyte distribution.** Most numerous in the epidermis of darker areas such as the nipples and areolae, and in the face, limbs, and mucous membranes. The number of melanocytes is about the same in all people. The difference in skin color comes from the amount of pigment those cells produce and transfer to keratinocytes.

**Melanin synthesis.** Made from the amino acid tyrosine in the presence of the enzyme tyrosinase, inside an organelle called the **melanosome**. Exposure to UV light increases the activity of that enzyme, which increases melanin production. That is a tan, and it is protective against further exposure.

## Hair

**Location.** Most skin surfaces. Exceptions: palms, fingertips, soles, and plantar surfaces of the feet. **Determinants of thickness and distribution.** Genetic and hormonal.

**Functions.** Limited protection from the sun, decreased heat loss from the scalp, eyelashes and eyebrows protecting the eyes from debris, nostril and ear hair protecting those structures from debris, and sensing light touch via hair root plexuses.

### Anatomy of a hair

**Composition.** Dead, keratinized epidermal cells. **Two parts.** The **hair root**, the portion penetrating into the dermis and sometimes the subcutaneous tissue, and the **hair shaft**, the portion above the surface of the skin. Both have the same three concentric layers.

1. **Medulla**, the inner layer, 2 to 3 rows of irregular cells containing pigment.
2. **Cortex**, the major part of the shaft, elongated cells.
3. **Cuticle**, the outermost, a single layer of thin, flat, heavily keratinized cells.

### The hair follicle

The follicle surrounds the root of the hair.

- **External root sheath**, a downward continuation of the epidermis.
- **Internal root sheath**, produced by the matrix, forming a cellular tubular sheath of epithelium. External plus internal root sheath equals the **epithelial root sheath**.
- **Dermal root sheath**, the dermis surrounding the hair follicle.

**The bulb** is the onion-shaped structure at the base of each follicle, surrounding the dermal root sheath. It contains the **papilla of the hair**, which is areolar connective tissue with many blood vessels, and the **hair matrix**, the germinal layer of cells arising from the stratum basale. The matrix is the site of cell division: it is responsible for the growth of existing hairs, the generation of new hairs when old ones are shed within the same follicle, and it gives rise to the cells of the internal root sheath.

### Structures associated with hair

- **Sebaceous oil glands**, secrete oil onto the hair.
- **Arrector pili muscles**, smooth muscle that pulls the hair shaft perpendicular, producing goose bumps.
- **Hair root plexus**, dendrites of neurons surrounding each follicle, detecting light touch.

### Types of hair and hair color

**Lanugo** is the very fine, non-pigmented downy hair covering the fetus. **Terminal hairs** are long, coarse, heavily pigmented hairs that replace lanugo just before birth on the eyebrows, eyelashes, and scalp. **Vellus hairs**, the peach fuzz, replace lanugo on the rest of the body. After puberty, under hormonal influence, some vellus hairs are replaced by terminal hairs, and the proportions differ between individuals.

**Hair color** is determined by the amount and type of melanin in the keratinized cells. Melanin is synthesized by melanocytes, scattered in the matrix of the bulb, and passes into the cells of the cortex and medulla. Dark hair is mostly eumelanin. Blond and red hair are mostly pheomelanin. Gray hair reflects a progressive decline in melanin production and contains few melanin granules. White hair lacks melanin entirely and accumulates air bubbles in the shaft.

## Skin Glands

| THE FOUR GLANDS COMPARED |                                                                                                             |                                                            |                                                                           |                                                                                                           |                                                             |
|--------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Gland                    | Distribution                                                                                                | Secretory portion                                          | Duct terminates                                                           | Secretion                                                                                                 | Onset                                                       |
| <b>Sebaceous (oil)</b>   | Largely lips, glans penis, labia minora, tarsal glands. Small on trunk and limbs. Absent on palms and soles | Dermis                                                     | Mostly connected to a hair follicle                                       | Sebum, a mixture of triglycerides, cholesterol, proteins, and inorganic salts                             | Relatively inactive through childhood, activated at puberty |
| <b>Eccrine sweat</b>     | Throughout most regions of the body, especially forehead, palms, and soles                                  | Mostly deep dermis, sometimes the upper subcutaneous layer | Surface of the epidermis                                                  | Perspiration: water, ions (sodium, chloride), urea, uric acid, ammonia, amino acids, glucose, lactic acid | Soon after birth                                            |
| <b>Apocrine sweat</b>    | Skin of axillae, groin, areolae, bearded regions of the face, clitoris and labia minora                     | Mostly deep dermis and upper subcutaneous layer            | Hair follicles                                                            | Perspiration with the same components as eccrine sweat plus lipids and proteins                           | Puberty                                                     |
| <b>Ceruminous</b>        | External auditory canal                                                                                     | Subcutaneous layer                                         | Surface of the external auditory canal, or into ducts of sebaceous glands | Cerumen, a waxy material                                                                                  | Soon after birth                                            |

## Nails

**Definition.** Plates of tightly packed, hard, dead keratinized epidermal cells forming a clear, solid covering over the dorsal surfaces of the distal portions of the digits.

| COMPONENTS OF A NAIL        |                                                                                                                                                                                                               |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Component                   | What it is                                                                                                                                                                                                    |
| <b>Nail body (plate)</b>    | The visible portion, comparable to the stratum corneum of the skin except the flattened keratinized cells fill with a harder keratin and are not shed. Mostly pink due to the vascularity of the tissue below |
| <b>Free edge</b>            | The part extending beyond the finger or toe. White because there are no capillaries beneath it                                                                                                                |
| <b>Nail root</b>            | The portion buried in a fold of skin, not visible                                                                                                                                                             |
| <b>Lunula</b>               | The little moon, a crescent-shaped area at the proximal end of the nail body. Vascular tissue does not show through because the epithelium there is thickened                                                 |
| <b>Hyponychium</b>          | Thickened stratum corneum below the free edge. Secures the nail to the fingertip                                                                                                                              |
| <b>Eponychium (cuticle)</b> | A narrow band of epidermis extending from and adhering to the margin of the nail wall. It is stratum corneum                                                                                                  |
| <b>Nail matrix</b>          | Epithelium proximal to the nail root, containing dividing cells that produce new nail cells                                                                                                                   |

**Functions.** Protect the ends of the digits, provide support and counter pressure to the palmar surface of the fingers which enhances touch perception and manipulation, and allow us to grasp and manipulate small objects, scratch, and groom.

### Thick Skin and Thin Skin

The greatest contributor to epidermal thickness is an increased number of layers in the stratum corneum, which arises in response to greater mechanical stress.

| STRUCTURAL DIFFERENCES                  |                                                                                  |                                                                                                    |
|-----------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Feature                                 | Thin (hairy) skin                                                                | Thick (hairless) skin                                                                              |
| <b>Distribution</b>                     | All parts of the body except palms, palmar surface of digits, and soles          | Palms, soles, palmar surface of digits                                                             |
| <b>Epidermal strata</b>                 | Stratum lucidum essentially lacking, thinner spinosum and corneum                | Stratum lucidum present, thicker spinosum and corneum                                              |
| <b>Epidermal ridges</b>                 | Lacking, because dermal papillae are poorly developed, fewer, and less organized | Present, because dermal papillae are well developed, more numerous, and organized in parallel rows |
| <b>Hair follicles and arrector pili</b> | Present                                                                          | Absent                                                                                             |
| <b>Sebaceous glands</b>                 | Present                                                                          | Absent                                                                                             |
| <b>Sudoriferous glands</b>              | Fewer                                                                            | More numerous                                                                                      |
| <b>Sensory receptors</b>                | Sparser                                                                          | Denser                                                                                             |

### Immune Protection, Sensation, Exchange

#### Two cell types with an immunological role

1. **Intraepidermal macrophages (Langerhans cells)**, alert the immune system to potentially harmful microbial invaders, recognizing and processing them.
2. **Macrophages in the dermis**, phagocytize bacteria and viruses that get past the intraepidermal macrophages of the epidermis.

#### Cutaneous sensations

Sensations that arise in the skin, including tactile sensations: touch, pressure, vibration, tickling, and thermal sensation.

**Pain** is the sensation indicating impending or actual tissue damage. The receptors include tactile discs of the epidermis, corpuscles of touch, and hair root plexuses around each follicle.

#### Excretion and absorption

The skin has a small role in both. **Excretion** is the elimination of substances from the body. **Absorption** is the passage of materials from the external surface inward, and although the amount is negligible, it does occur.

**What crosses.** Lipid-soluble materials: the fat-soluble vitamins A, D, E, and K, certain drugs, gases such as oxygen and carbon dioxide, toxic materials, salts of heavy metals such as lead, mercury, and arsenic, and the oils of poison ivy and poison oak. Steroid drugs such as cortisone are lipid soluble and move into the papillary region of the dermis, where their anti-inflammatory effect includes inhibiting histamine production by mast cells. Other drugs are delivered through patches, nitroglycerin and fentanyl among them.

## Synthesis of vitamin D

Only a small amount of exposure, 10 to 15 minutes twice a week, is necessary for vitamin D synthesis.

## Clinical Correlations

### Skin grafts

A **skin graft** is the transfer of a patch of healthy skin from a donor site to cover a wound. New skin cannot regenerate if the injury has damaged a large area of the stratum basale and its stem cells, which is exactly why the depth of a burn determines the treatment.

**Benefits.** Protects against fluid loss and infection, promotes tissue healing, reduces scar formation, prevents loss of function, and improves cosmetic outcome.

**Complication.** Tissue rejection. To minimize the risk, transplanted skin is taken from the same person, an **autograft**, or from an identical twin, an **isograft**. In severe burns where taking an autograft would cause more harm than good, **autologous skin transplantation** is used: small amounts of the person's epidermis are removed, keratinocytes are cultured in the lab and grown into thin sheets of skin, and those sheets are transplanted.

### Albinism and vitiligo

**Albinism** is the inherited inability to produce melanin. Most affected individuals have melanocytes but cannot synthesize tyrosinase, so melanin is missing from skin, eyes, and hair. Manifestations include problems with night vision and a tendency for the skin to burn with sun overexposure.

**Vitiligo** is the partial or complete loss of melanocytes from patches of skin, producing irregular white spots.

### Skin color as a diagnostic clue

| FOUR COLOR CHANGES AND WHAT THEY SUGGEST |                           |                                                                                                                              |
|------------------------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Finding                                  | Appearance                | What it suggests                                                                                                             |
| <b>Cyanosis</b>                          | Blue                      | Blood is not picking up adequate oxygen from the lungs. Most apparent in mucous membranes, nail beds, and skin               |
| <b>Jaundice</b>                          | Yellowish skin and sclera | A build-up of the yellow pigment bilirubin, likely indicating liver disease                                                  |
| <b>Erythema</b>                          | Redness                   | Engorgement of dermal capillaries with blood, from skin injury, heat exposure, infection, inflammation, or allergic reaction |
| <b>Pallor</b>                            | Paleness                  | May occur in conditions such as shock or anemia                                                                              |

These changes can be difficult to assess in individuals with darker skin. Examining the nail beds, gums, mucous membranes, and conjunctiva is the reliable approach.

### Thermoregulation

| THE SKIN'S TWO-DIRECTION RESPONSE |                                                                                     |                                                               |                                                  |
|-----------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------|
| When the body is                  | Blood vessels                                                                       | Sweat                                                         | Other                                            |
| <b>Too warm</b>                   | Vasodilation brings warm blood to the surface so heat is lost to the environment    | Sweat production increases, and evaporation cools the surface | Goal is to shed heat                             |
| <b>Too cold</b>                   | Vasoconstriction keeps blood away from the surface so heat is conserved in the core | Sweat production is reduced                                   | Shivering generates heat through muscle activity |

**THE SKIN LAYERS MAP THE DEPTH OF A BURN**

Burn classification is anatomy applied. A superficial burn involves the epidermis only, and it heals because the stratum basale is intact and its stem cells can repopulate the layers above. A partial-thickness burn reaches into the dermis, blisters because the dermal-epidermal junction fails, and still heals from surviving follicles and glands, since those structures are lined with epidermal cells that live deep in the dermis. A full-thickness burn destroys the dermis and its appendages, so there is no source of new epidermis left within the wound, and grafting becomes necessary. Read every burn question by asking one thing first: how deep did it go, and is the stratum basale still there.

PART B

# Pre-work.

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Four packets, in the order they are due. Work them in sequence: read the Part A chapter, organize it into mini visuals, draw it from memory, apply it with your notes closed, then retrieve later in the week by redrawing from memory. Context before content. This is not a night-before task.

**1**    **Intro to Anatomy**

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**2**    **Cell Anatomy**

---

**3**    **Tissues and Histology**

---

**4**    **Integumentary System**

---

THESE PAGES PRINT LANDSCAPE. TURN THE PACKET SIDEWAYS.



## Pre-work · Intro to Anatomy

Intro to Anatomy. Read the Part A chapters first, then work through this in order. When you walk into class you should be ready to use this material, not hearing it for the first time.

Module 1 · 1 of 4

Bring this to class

### How to work this pre-work

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

- 1 Start with the reading.**  
Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.
- 2 Organize what you read.**  
In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.
- 3 Draw it.**  
In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.
- 4 Apply it.**  
Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.
- 5 Come back to it later in the week**  
, not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

**Here is why the order works.** Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

#### PANEL 1 OF 3

### Organize it

Turn the two intro chapters into mini visuals. Seven chunks, each in the shape that fits it.

#### PANEL 2 OF 3

### Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

#### PANEL 3 OF 3

### Apply it

First pass with everything closed. Then open the notes in a second color and mark what you missed.

### YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

## Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

### 1 LADDER

#### The six levels of structural organization

Six rungs, simplest at the bottom. Write one real example beside each rung, not the one from the notes.

### 2 SPLIT

#### The seven directional pairs

One split per pair. Pair on the branches, and underneath, the one sentence that proves you can use it. Star the two you are least sure of.

### HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

### WORKED EXAMPLE

The elbow is \_\_\_\_\_ to the wrist.

**Answer.** Proximal.

**Reasoning.** Both structures are on the same limb, so the proximal and distal pair is available. Proximal means nearer the point of attachment to the trunk. The elbow attaches nearer the shoulder than the wrist does, so the elbow is proximal. Note the wrong answer you could reach for: the elbow is also superior to the wrist in anatomical position, but that is the weaker answer, because the question is about a limb and the limb-specific pair is the precise one.

## Set A · Directional relationships

1. The sternum is \_\_\_\_\_ to the vertebral column.
2. The lungs are \_\_\_\_\_ to the diaphragm.
3. The thumb is \_\_\_\_\_ to the little finger, in anatomical position.
4. The skin of the thigh is \_\_\_\_\_ to the femur.
5. A stroke in the left cerebral hemisphere causes weakness on the right side of the body. That effect is \_\_\_\_\_.
6. Explain in one sentence why it is wrong to say the nose is proximal to the chin.

---

---

## Set B · Planes and sections

1. A scan of the head shows both eyes and both ears in the same image, with the top of the skull missing. Which plane produced it, and how do you know?

---

---

### 3 GRID

#### Planes crossed with what they divide

Planes down the side, the two parts each produces across the top. Six rows. Draw the cut line on a stick figure in one box.

### 4 HUB

#### Body cavities

Ventral and dorsal as two hubs. Spoke out to each subdivision, then to what it contains. Mark the diaphragm on the spoke it actually sits on.

### 5 CHAIN

#### Serous membrane, wall to organ

Body wall, arrow, parietal layer, arrow, cavity with fluid, arrow, visceral layer, arrow, organ. Then write the chain three times, once each for pericardium, pleura, peritoneum.

### 6 HUB

#### The mediastinum

Borders as spokes on one side, contents as spokes on the other. Five borders, five contents.

## DRAWING 1

### Anatomical position, with the planes on it

#### BRAIN DUMP FIRST

Two minutes. Anatomical position and every plane you can name, with what each one separates.

Draw a full figure in anatomical position. Mark the five features that define it: feet flat, arms at the sides, palms forward, head and eyes forward, body erect. Then cut the same figure with the sagittal, frontal, and transverse planes. Label each, and write the two portions each one creates.

*Room to draw this is on the drawing pages at the end of this pre-work.*

In your notebook, head the page **M1-1 Anatomical position, with the planes on it** so you can find it later. Correct it in a second color.

#### YOU KNOW IT WHEN

You can use midsagittal and parasagittal on one figure and say the difference in a sentence.

2. A cut separates the body into anterior and posterior portions. Name the plane.

3. A section through the arm shows a single round bone surrounded by a ring of muscle. Which plane, and how do you know?

4. You want an image showing the relationship between the front of the heart and the sternum. Which plane serves you best, and why?

## Set C - Cavities and membranes

1. A stab wound opens the pleural cavity. Predict what happens to the lung on that side, and say why.

2. A patient has a ruptured appendix. Name the membrane involved, the condition it produces, and why the problem does not stay local.

3. A surgeon plans to reach the right kidney. Is the approach through the peritoneal cavity or behind it? Name the term and one other structure in the same space.

4. A friction rub is heard over the heart. Name the two layers rubbing, the fluid that should be preventing it, and the condition.

## 7 GRID

### Quadrants and regions on one abdomen

Draw the abdomen once. Quadrant lines in one color, the nine-region lines in a second. Label the organs where they sit. One drawing does both systems.

#### YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

## DRAWING 2

### The cavities, lateral view

#### BRAIN DUMP FIRST

Two minutes. Every cavity, what is in it, and what separates it from the one next door.

Draw the body in profile. Outline the dorsal cavity and the ventral cavity, then subdivide each. Shade the thoracic and abdominopelvic cavities differently and draw the diaphragm between them. Put the mediastinum in, and shade the retroperitoneal space behind the peritoneum.

*Room to draw this is on the drawing pages at the end of this pre-work.*

In your notebook, head the page **M1-2 The cavities, lateral view** so you can find it later. Correct it in a second color.

#### YOU KNOW IT WHEN

You can point at any organ on your drawing and say which cavity it sits in and which membrane wraps it.

## Set D · Locate it

1. A patient reports pain in the right lower quadrant. Name two organs that could be responsible.  
\_\_\_\_\_
2. Give the regional term for the front of the elbow, the back of the knee, and the sole of the foot.  
\_\_\_\_\_
3. Sort as axial or appendicular: brachial, thoracic, popliteal, vertebral, carpal.  
\_\_\_\_\_  
\_\_\_\_\_

#### YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

### DRAWING 3

#### Serous membrane, the fist and the balloon

##### BRAIN DUMP FIRST

One minute. Parietal, visceral, cavity, fluid.  
What each one touches.

Push a fist into a partly inflated balloon and draw that. The layer against your fist is visceral, the outer layer is parietal, the space between holds the fluid. Label all four. Then draw what happens to the space when it fills with air, and when it fills with fluid.

*Room to draw this is on the drawing pages at the end of this pre-work.*

In your notebook, head the page **M1-3 Serous membrane, the fist and the balloon** so you can find it later. Correct it in a second color.

##### YOU KNOW IT WHEN

You can explain a collapsed lung and a friction rub using only this drawing.

## Mini visual sheet

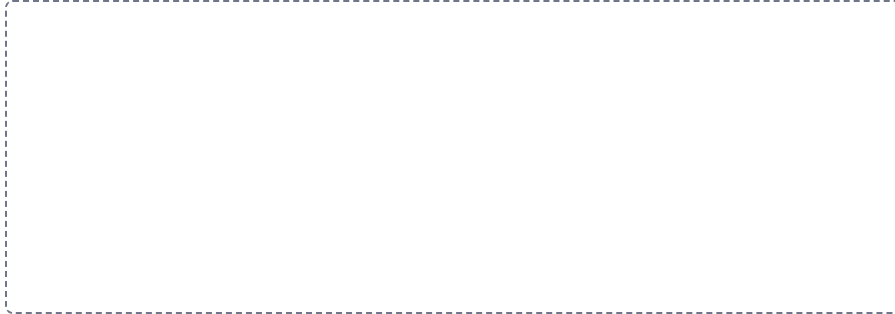
Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

1

### The six levels of structural organization

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

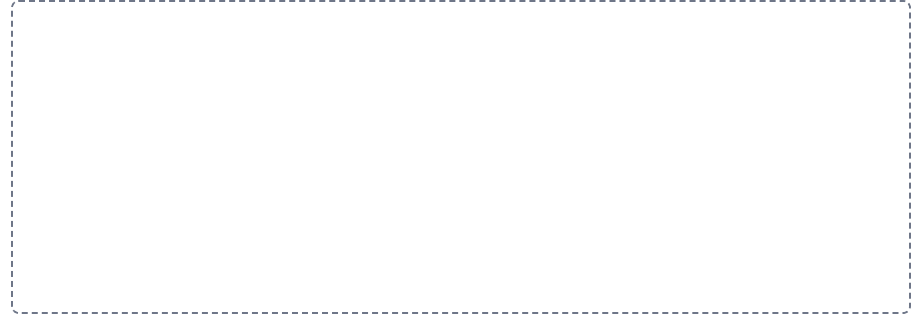


2

### The seven directional pairs

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.



3

### Planes crossed with what they divide

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.



4

### Body cavities

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.



5

### Serous membrane, wall to organ

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

6

### The mediastinum

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

7

### Quadrants and regions on one abdomen

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

## Drawing pages

The drawings from Panel 2, with room to draw them. Brain dump first with everything closed, then draw, then correct yourself in a second color.

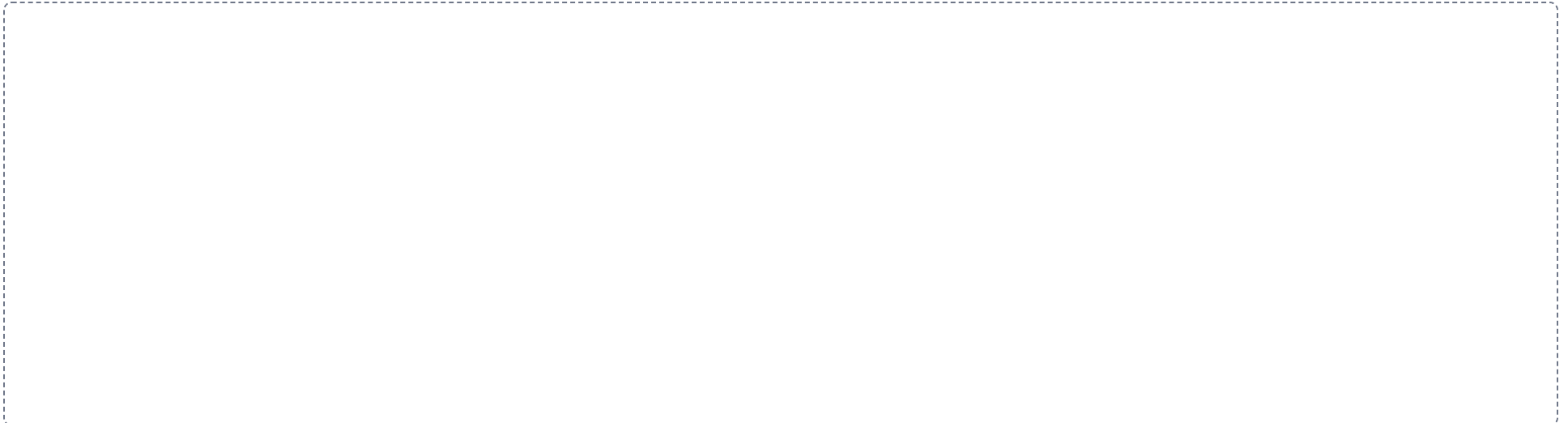
### DRAWING 1

**Anatomical position, with the planes on it**



### DRAWING 2

**The cavities, lateral view**





DRAWING 3

Serous membrane, the fist and the balloon



## Pre-work · Cell Anatomy

Cell Anatomy. Read the Part A chapter first, then work this in order. You come to class ready to use this, not to hear it for the first time.

Module 1 · 2 of 4

Bring this to class

### How to work this pre-work

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- 5 Come back to it later in the week**  
, not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

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#### PANEL 1 OF 3

### Organize it

Turn the cell chapter into mini visuals. Eight chunks, eight small visuals, each in the shape that fits it.

#### PANEL 2 OF 3

### Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

#### PANEL 3 OF 3

### Apply it

First pass with everything closed. Then open the notes in a second color and mark what you missed.

### YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

## Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

### 1 HUB

#### The generalized cell

One hub, three spokes: plasma membrane, nucleus, cytoplasm. Hang one job off each spoke. This is the frame everything else attaches to.

### 2 GRID

#### Permeability

Semipermeable, permeable, impermeable down the side. Definition and one example across the top. Three rows, and the examples must be yours.

### HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

### WORKED EXAMPLE

**A drug destroys the enzymes inside lysosomes. Predict what accumulates and where.**

**Answer.** Undigested macromolecules and worn organelles build up inside the cell.

**Reasoning.** Work from structure to consequence. The lysosome is the digestive organelle, so its job is breaking down macromolecules, damaged organelles, and pathogens. Remove the enzymes and nothing else takes over that job, so the material has nowhere to go and stays inside the cell. Cells with the highest turnover show it first. This is the mechanism behind Tay-Sachs, where one missing lysosomal enzyme lets lipid accumulate in nerve cells until the cell fails.

## Set A - Structure to job

1. The organelle that assembles ribosomal subunits is the \_\_\_\_\_.
2. A cell that secretes large amounts of protein would be rich in \_\_\_\_\_.
3. A cell that detoxifies drugs would be rich in \_\_\_\_\_.
4. The organelle with its own DNA and ribosomes is the \_\_\_\_\_.
5. Name one cell type that is multinucleated and one that is anucleated.  
\_\_\_\_\_

## Set B - Predict the failure

1. A mutation stops the cell producing glycoproteins. Predict two things the cell can no longer do well, and say why.  
\_\_\_\_\_  
\_\_\_\_\_
2. A cell cannot make enough ATP. Name the organelle involved, and name the three tissues that would show it first. Explain your choice.  
\_\_\_\_\_

### 3 HUB

#### Inside the nucleus

Nucleus at the center. Spoke to envelope, pores, chromatin, chromosomes, nucleoplasm, nucleolus. One job on each spoke.

### 4 SPLIT

#### Cytoplasm versus cytosol

One split. The trap is that one contains the other. Write the containing relationship on the branch, not just two definitions.

### 5 CHAIN

#### The secretory pathway

Nucleus, arrow, rough ER, arrow, vesicle, arrow, Golgi, arrow, vesicle, arrow, plasma membrane. Then write what happens to the protein at each arrow.

### 6 GRID

#### Rough ER against smooth ER

The two down the side. Structure, function, and what it sends onward across the top.

## DRAWING 1

### The generalized cell, labelled

#### BRAIN DUMP FIRST

Two minutes. Every organelle you can name and one job for each.

Draw one cell large enough to label. Include the plasma membrane, nucleus with nucleolus, rough and smooth ER, Golgi, mitochondria, lysosomes, peroxisomes, ribosomes, cytoskeleton, centrioles. Label every one. Do not draw a diagram you have seen, draw the one you remember.

*Room to draw this is on the drawing pages at the end of this pre-work.*

In your notebook, head the page **M1-4 The generalized cell, labelled** so you can find it later. Correct it in a second color.

#### YOU KNOW IT WHEN

You can point at any organelle on your drawing and name what fails if it stops working.

3. The cilia lining the airway stop moving. Predict the clinical picture, and say which microtubule arrangement has failed.

4. Cholesterol is removed from the membrane. Predict what happens to the membrane as temperature changes.

### Set C - Tell it apart

1. A student says cytoplasm and cytosol are the same thing. Correct them in one sentence.

2. Free ribosomes and bound ribosomes make different proteins. Say what decides which is used.

3. Chromatin and chromosomes are the same material. Say what changes between them and when.

#### YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

## 7 HUB

### The cytoskeleton

Three spokes: microfilaments, intermediate filaments, microtubules. Protein on one line, job on the next. Circle the one that builds cilia.

## 8 SPLIT

### 9 plus 0 against 9 plus 2

Centrioles on one branch, cilia and flagella on the other. Write the arrangement and, underneath, the one word that separates them: movement.

#### YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

## DRAWING 2

### The plasma membrane, close up

#### BRAIN DUMP FIRST

One minute. Every component of the membrane and where it sits.

Draw a section of bilayer wide enough to show detail. Mark the hydrophilic heads and hydrophobic tails and which way each faces. Add an integral protein spanning the layer, a peripheral protein on the surface, cholesterol between the tails, and a glycoprotein with its sugar chain facing out. Label which side is extracellular.

*Room to draw this is on the drawing pages at the end of this pre-work.*

In your notebook, head the page **M1-5 The plasma membrane, close up** so you can find it later. Correct it in a second color.

#### YOU KNOW IT WHEN

You can explain why a lipid-soluble drug crosses this and a charged ion does not, using only your drawing.

### DRAWING 3

#### Trace the pathway on your cell

##### BRAIN DUMP FIRST

One minute. The route a secreted protein takes, in order.

Go back to your first drawing. In a second color, draw the route a secreted protein takes from the gene to the outside of the cell. Number each step. Mark where the protein is made, where it is modified, and how it moves between stations.

*Room to draw this is on the drawing pages at the end of this pre-work.*

In your notebook, head the page **M1-6 Trace the pathway on your cell** so you can find it later. Correct it in a second color.

##### YOU KNOW IT WHEN

You can tell the story out loud without looking, and your numbers are in the right order.

## Mini visual sheet

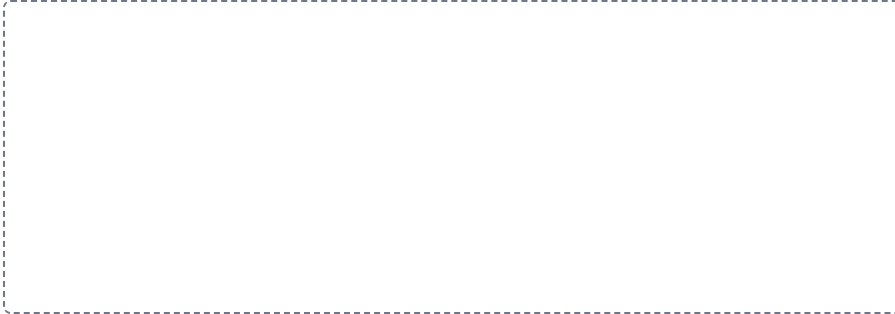
Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

1

### The generalized cell

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

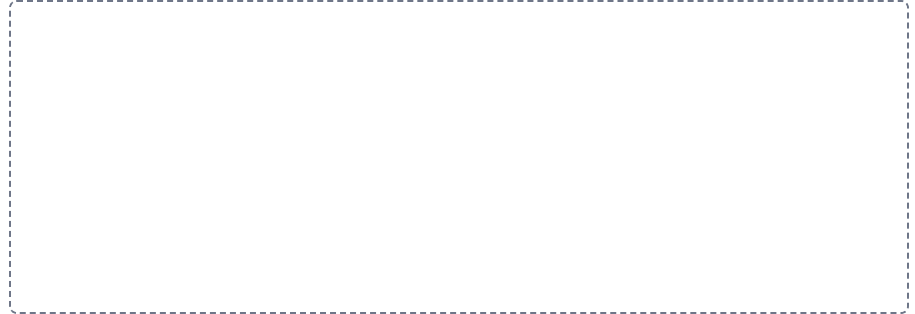


2

### Permeability

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.



3

### Inside the nucleus

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.



4

### Cytoplasm versus cytosol

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.



5

### The secretory pathway

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

6

### Rough ER against smooth ER

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

7

### The cytoskeleton

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

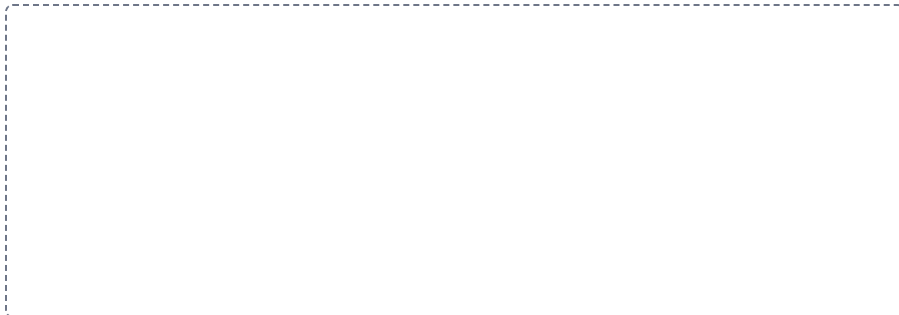


8

### 9 plus 0 against 9 plus 2

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.



## Drawing pages

The drawings from Panel 2, with room to draw them. Brain dump first with everything closed, then draw, then correct yourself in a second color.

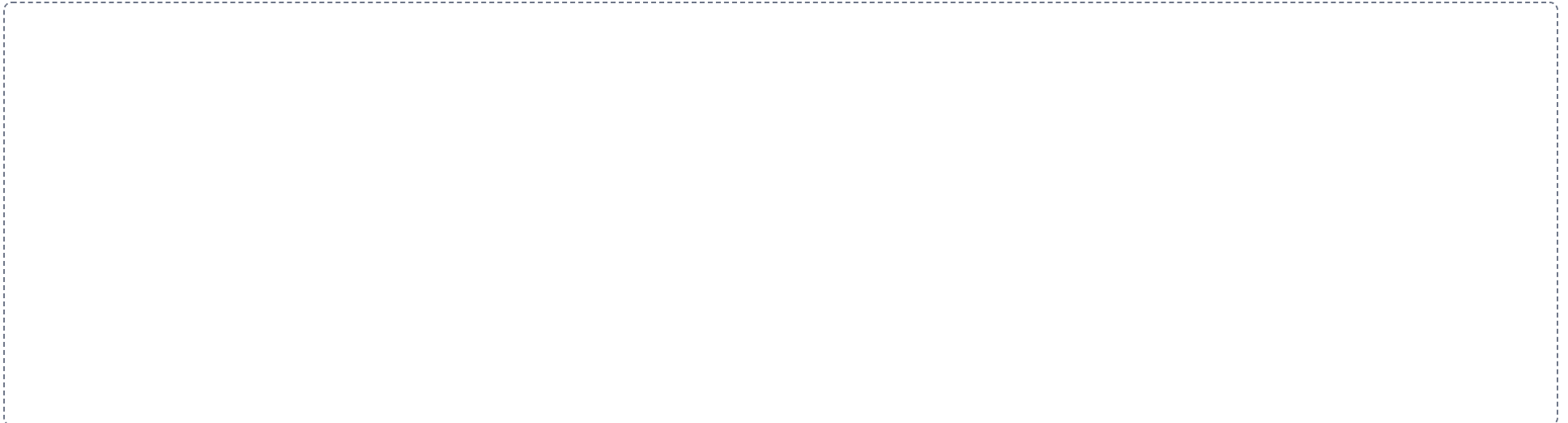
### DRAWING 1

**The generalized cell, labelled**



### DRAWING 2

**The plasma membrane, close up**



DRAWING 3

Trace the pathway on your cell



## Pre-work · Tissues and Histology

Tissues and Histology. Read the Part A chapter first, then work this in order. Most of this packet is pattern recognition, and pattern recognition is built by drawing, not by reading.

[Module 1 · 3 of 4](#)[Bring this to class](#)

### How to work this pre-work

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

- 1 Start with the reading.**  
Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.
- 2 Organize what you read.**  
In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.
- 3 Draw it.**  
In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.
- 4 Apply it.**  
Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.
- 5 Come back to it later in the week**  
, not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

**Here is why the order works.** Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

#### PANEL 1 OF 3

### Organize it

Turn the histology chapter into mini visuals. Eight chunks, and the two grids do most of the work.

#### PANEL 2 OF 3

### Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

#### PANEL 3 OF 3

### Apply it

First pass with everything closed. Then open the notes in a second color and mark what you missed.

### YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

### Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

#### 1 HUB

### The four tissue types

One hub, four spokes. One job and one location per spoke. Everything else in this chapter hangs off this frame.

### HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

### WORKED EXAMPLE

**A slide shows several layers of cells, and the cells at the free surface are flat. What is it, and where would you expect to find it?**

**Answer.** Stratified squamous epithelium. Skin if keratinized, otherwise esophagus, mouth, or vagina.

**Reasoning.** Name the layers first, then the shape at the free surface. Several layers means stratified. The cells at the top are flat, so squamous. That is the full name, and the naming rule always reads the apical layer, never the basal one. Then ask what the tissue is for: many layers that shed from the top means it survives abrasion, so look for it wherever something rubs.

### Set A - Name it from the description

1. A single layer of tall cells with goblet cells and microvilli. Name it and give one location.  
\_\_\_\_\_
2. Several layers where all cells touch the basement membrane but the nuclei sit at different heights. Name it.  
\_\_\_\_\_
3. Cells that change shape from dome to flat as the organ fills. Name it and the organ.  
\_\_\_\_\_
4. A single layer of flat cells one cell thick. Name it and say what that thinness makes possible.  
\_\_\_\_\_

### Set B - Junctions and consequences

1. A condition destroys desmosomes in the epidermis. Predict what happens to the skin and say why.  
\_\_\_\_\_  
\_\_\_\_\_

## 2 GRID

### Epithelium, layers crossed with shape

Simple and stratified down the side. Squamous, cuboidal, columnar across the top. Fill each box with a location. Add pseudostratified and transitional as two extra rows and say why they do not fit the grid cleanly.

## 3 GRID

### The five cell junctions

Junction down the side. Protein, job, and one location across the top. Five rows. Star the two you will confuse.

## 4 SPLIT

### Epithelium against connective tissue

One split, three branches on each side: cells and matrix, blood supply, location. The contrast is the point.

## DRAWING 1

### The eight epithelia, thumbnails

#### BRAIN DUMP FIRST

Two minutes. Every epithelium you can name, with its shape.

Divide the box into eight small frames. Sketch each epithelium as it looks on a slide: the basement membrane as a line, the cells above it in the right shape and number of layers. Add cilia, microvilli, goblet cells, and keratin where they belong. Label each and write one location under it. Small and fast, not beautiful.

*Room to draw this is on the drawing pages at the end of this pre-work.*

In your notebook, head the page **M1-7 The eight epithelia, thumbnails** so you can find it later. Correct it in a second color.

#### YOU KNOW IT WHEN

Someone shows you a slide and you can name it from the shape of the top layer and the number of layers, without reading a caption.

2. A toxin opens the tight junctions in the intestinal lining. Predict what can now cross that could not before.

3. Cardiac muscle relies on two junction types at the intercalated disc. Name both and say what each contributes.

## Set C - Connective tissue and glands

1. A tendon and a skin wound are injured on the same day. Predict which heals faster and explain using vascularity.

2. A gland destroys its own cells to release its product. Name the mode and one gland that uses it.

3. A patient has an allergic reaction with swelling and redness. Name the connective tissue cell responsible and the substance it released.

4. Sort by matrix, softest to hardest: bone, blood, cartilage, areolar tissue. Say what changes across that order.

#### YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

## 5 HUB

### What connective tissue is made of

Hub is extracellular matrix plus cells. Spoke to ground substance and to the three fibers, and separately to the six cell types. One job each.

## 6 LADDER

### Blasts and cytes

Two rungs. Blasts on the lower rung building, cytes above maintaining. Write three examples of each pair beside the rungs.

## 7 CHAIN

### How a gland releases its product

Three short chains, one per mode. Merocrine, apocrine, holocrine. Draw what happens to the cell at the end of each chain. That difference is the whole classification.

## 8 GRID

### Cartilage and bone

Hyaline, elastic, fibrocartilage, compact bone down the side. Matrix, cells, perichondrium yes or no, and location across the top.

## DRAWING 2

### The five junctions on two cells

#### BRAIN DUMP FIRST

One minute. The five junctions and what each one attaches to.

Draw two adjacent cells sitting on a basement membrane. Put all five junctions on them in the right places: tight junction near the apex, adherens belt below it, desmosomes between the cells, hemidesmosomes at the basement membrane, gap junctions as channels. Label the protein at each.

*Room to draw this is on the drawing pages at the end of this pre-work.*

In your notebook, head the page **M1-8 The five junctions on two cells** so you can find it later. Correct it in a second color.

#### YOU KNOW IT WHEN

You can explain a blistering condition by pointing to which junction failed.

#### YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

### DRAWING 3

#### A connective tissue field

##### BRAIN DUMP FIRST

One minute. Every fiber and every cell type you can name.

Draw a field of loose areolar tissue. Show collagen fibers as thick wavy bundles, elastic fibers as thin branching threads, and ground substance as the space between. Add a fibroblast, a macrophage, a mast cell, an adipocyte, and a plasma cell, and label each. Then in a second color, redraw the same field as dense regular tissue and say what changed.

*Room to draw this is on the drawing pages at the end of this pre-work.*

In your notebook, head the page **M1-9 A connective tissue field** so you can find it later. Correct it in a second color.

##### YOU KNOW IT WHEN

You can look at your two versions and say why one heals fast and the other heals slowly.



## Mini visual sheet

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

1

### The four tissue types

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

2

### Epithelium, layers crossed with shape

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

3

### The five cell junctions

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

4

### Epithelium against connective tissue

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

5

### What connective tissue is made of

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

6

### Blasts and cytes

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

7

### How a gland releases its product

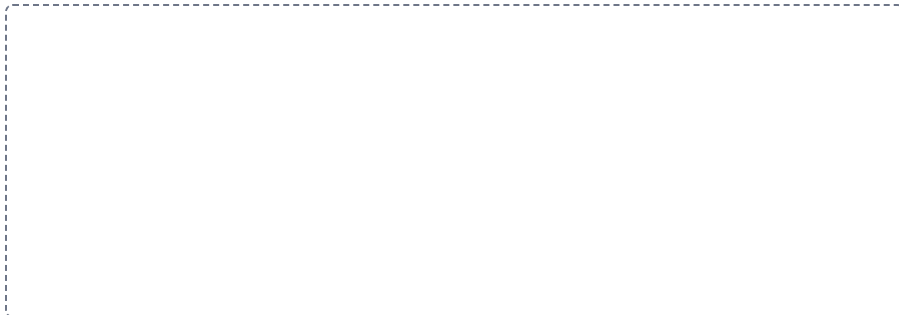
CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

## Cartilage and bone

### GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.



## Drawing pages

The drawings from Panel 2, with room to draw them. Brain dump first with everything closed, then draw, then correct yourself in a second color.

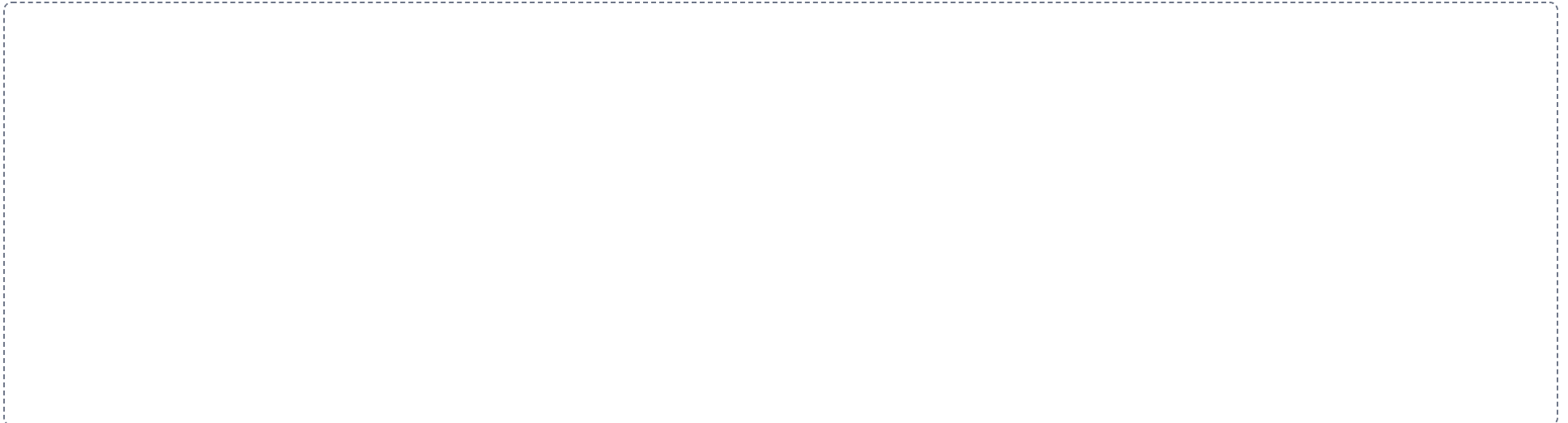
### DRAWING 1

#### The eight epithelia, thumbnails



### DRAWING 2

#### The five junctions on two cells



DRAWING 3

A connective tissue field



## Pre-work · Integumentary System

Integumentary System. Read the Part A chapter first, then work this in order. This is the first organ where all four tissue types show up together, so it is where the module gets used rather than listed.

Module 1 · 4 of 4

Bring this to class

### How to work this pre-work

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

- 1 Start with the reading.**  
Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.
- 2 Organize what you read.**  
In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.
- 3 Draw it.**  
In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.
- 4 Apply it.**  
Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.
- 5 Come back to it later in the week**  
, not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

**Here is why the order works.** Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

#### PANEL 1 OF 3

### Organize it

Turn the integumentary chapter into mini visuals. Seven chunks, and the ladder is the one you will use on every exam.

#### PANEL 2 OF 3

### Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

#### PANEL 3 OF 3

### Apply it

First pass with everything closed. Then open the notes in a second color and mark what you missed.

### YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

### Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

#### 1 LADDER

#### The five strata

Five rungs, basale at the bottom. Beside each rung write what the keratinocyte is doing at that level. The mnemonic goes in the margin, not on the ladder.

### HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

### WORKED EXAMPLE

**A burn destroys the epidermis and the upper dermis but leaves hair follicles and glands intact. Predict whether it regenerates, and say why.**

**Answer.** Yes, it regenerates, from the follicles and glands.

**Reasoning.** Ask one question of every burn: is there a source of new epidermal cells left inside the wound. The stratum basale is gone at the surface, but hair follicles and glands are lined with epidermal cells and they sit deep in the dermis, so they survive. New epithelium spreads outward from each surviving structure until the sheets meet. A full-thickness burn destroys those too, which is why it cannot heal from the middle and needs grafting.

### Set A · Layers and cells

1. Name the strata in order, deep to superficial, for thick skin.  
\_\_\_\_\_
2. Which stratum is missing in thin skin, and where on the body would you find thick skin?  
\_\_\_\_\_
3. The cell that alerts the immune system to an invader in the epidermis is the \_\_\_\_\_.
4. Which epidermal cell is least numerous, where does it sit, and what does it contact?  
\_\_\_\_\_

### Set B · Color and pigment

1. Two people have very different skin color but the same number of melanocytes. Explain the difference.  
\_\_\_\_\_  
\_\_\_\_\_

## 2 HUB

### The four cells of the epidermis

One hub, four spokes: keratinocyte, melanocyte, intraepidermal macrophage, Merkel cell. Percentage on one line, job on the next. Mark which stratum each lives in.

## 3 CHAIN

### Melanin transfer

Melanocyte, arrow, projection between keratinocytes, arrow, granules enter the keratinocyte, arrow, granules cluster over the nucleus. Draw the last step, do not just write it. That veil is the whole point.

## 4 SPLIT

### Papillary against reticular dermis

One split. Fraction of the dermis, fibers, and what lives in each on the branches. Add the three structures inside a dermal papilla.

## DRAWING 1

### The skin in cross section

#### BRAIN DUMP FIRST

Two minutes. Every layer and every structure in the skin.

Draw the full thickness: epidermis, dermis with papillary and reticular regions, and hypodermis. Add a hair follicle running down, a sebaceous gland on it, an arrector pili muscle, an eccrine gland with its duct reaching the surface, dermal papillae with capillary loops and Meissner corpuscles, and a lamellated corpuscle in the hypodermis. Label everything.

*Room to draw this is on the drawing pages at the end of this pre-work.*

In your notebook, head the page **M1-10 The skin in cross section** so you can find it later. Correct it in a second color.

#### YOU KNOW IT WHEN

You can draw a horizontal line across your own drawing at any depth and say what a burn to that depth would destroy.

2. A patient appears yellow in the skin and the sclera. Name the finding, the pigment, and the organ you would suspect.

---

---

3. A patient with dark skin is suspected of being cyanotic. Say where you would look instead, and why.

---

---

4. Explain why a tan is protective and why melanocytes are still damaged by the same exposure.

---

---

### Set C · Structures and consequences

1. A gland opens onto a hair follicle in the axilla and switches on at puberty. Name it.

2. Damage to the nail matrix. Predict the effect on the nail and say why.

---

---

3. A patient loses the ability to sweat over a large area. Predict the consequence in hot weather and name the function lost.

---

---

4. Explain why a steroid cream works on inflamed skin but a large protein would not cross at all.

---

---



## 5 GRID

### The four skin glands

Sebaceous, eccrine, apocrine, ceruminous down the side. Where they are, what they secrete, where the duct opens, and when they switch on across the top.

## 6 HUB

### The nail

Nail at the center. Spoke to body, free edge, root, lunula, hyponychium, eponychium, matrix. One line each on what it is or does.

## 7 CHAIN

### Thermoregulation, both directions

Two chains from one starting point. Too warm goes one way, too cold the other. Put vessels and sweat on both chains and shivering on one.

#### YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

## DRAWING 2

### The five strata, cell by cell

#### BRAIN DUMP FIRST

One minute. The strata in order and what changes between them.

Zoom in on the epidermis alone. Draw the five strata stacked, with the cells changing shape as they rise: cuboidal and dividing at the base, spiny in the middle, granular above, clear, then flat dead squames at the top. Mark where the nucleus disappears and where lamellar granules release their sealant. Mark which layer is missing in thin skin.

*Room to draw this is on the drawing pages at the end of this pre-work.*

In your notebook, head the page **M1-11 The five strata, cell by cell** so you can find it later. Correct it in a second color.

#### YOU KNOW IT WHEN

You can explain why the epidermis survives without a blood supply, using only your drawing.

#### YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

## DRAWING 3

### Hair follicle and bulb

#### BRAIN DUMP FIRST

One minute. The parts of a hair and everything attached to the follicle.

Draw one follicle from the surface down to the bulb. Show the shaft above the skin and the root below, the three concentric layers, the external and internal root sheaths, the dermal root sheath, and at the base the bulb with the papilla and the matrix. Attach the sebaceous gland, the arrector pili, and the hair root plexus.

*Room to draw this is on the drawing pages at the end of this pre-work.*

In your notebook, head the page **M1-12 Hair follicle and bulb** so you can find it later. Correct it in a second color.

#### YOU KNOW IT WHEN

You can say where the hair actually grows from and why cutting it does not change that.

## Mini visual sheet

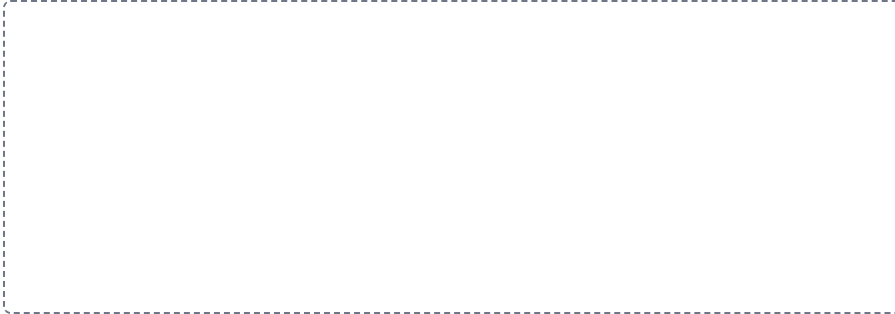
Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

1

### The five strata

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

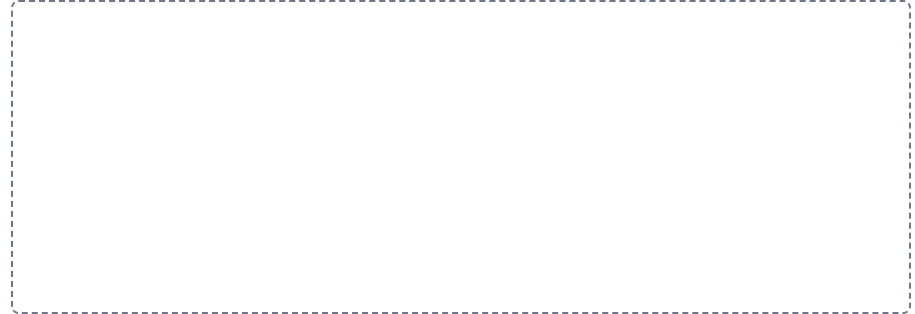


2

### The four cells of the epidermis

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.



3

### Melanin transfer

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.



4

### Papillary against reticular dermis

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

5

### The four skin glands

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

6

### The nail

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

7

## Thermoregulation, both directions

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

## Drawing pages

The drawings from Panel 2, with room to draw them. Brain dump first with everything closed, then draw, then correct yourself in a second color.

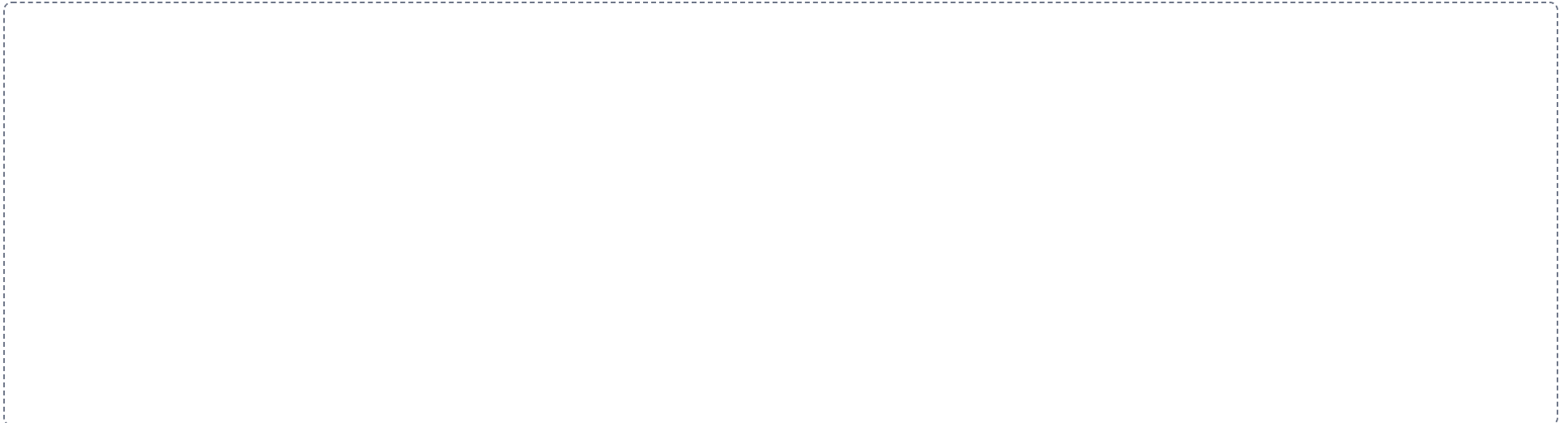
### DRAWING 1

#### The skin in cross section



### DRAWING 2

#### The five strata, cell by cell



DRAWING 3

Hair follicle and bulb



## PART C

# Lab Structure List.

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Part C of the Module 1 packet. These are the lab structures and concepts to learn for the lab portion of Exam 1. Treat it as the master list, and cross reference it against every lab worksheet. Be prepared to identify structures on models, images, the cadaver, and slides.

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**1 Introduction to Anatomy**

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**2 Epithelial Tissues**

---

**3 Connective Tissues**

---

**4 Muscle Tissues**

---

**5 Nerve Tissues**

---

**6 Blood Vessels and Formed Elements**

---

**7 Integumentary System**

---



## CHAPTER 1

# Introduction to Anatomy

Body landmarks, orientation, planes, cavities, membranes, and the microscope. Everything here is testable on models, images, the cadaver, and slides.

1. **All anterior and posterior body regional landmarks.** Know the term, and be able to point to it on a model and on yourself.

## Anterior regions

| ANTERIOR REGIONAL TERMS |                                   |                       |                              |
|-------------------------|-----------------------------------|-----------------------|------------------------------|
| Term                    | Region                            | Term                  | Region                       |
| Frontal                 | Forehead                          | Inguinal              | Groin                        |
| Cranial                 | Skull or head                     | Pubic                 | Area near the genital region |
| Facial                  | Face                              | Brachial              | Upper arm                    |
| Orbital or ocular       | Eye region                        | Antebrachial          | Forearm                      |
| Buccal                  | Cheek                             | Antecubital           | Front of the elbow           |
| Otic                    | Ear region                        | Carpal                | Wrist                        |
| Nasal                   | Nose                              | Pollex                | Thumb                        |
| Oral                    | Mouth region                      | Palmar                | Palm of the hand             |
| Mental                  | Chin                              | Digital or phalangeal | Fingers or toes              |
| Cervical                | Neck region                       | Patellar              | Kneecap                      |
| Thoracic                | Chest area                        | Crural                | Leg, the shin area           |
| Sternal                 | Breastbone                        | Tarsal                | Ankle                        |
| Mammary                 | Breast region                     | Pedal                 | Foot                         |
| Abdominal               | Below the chest, above the pelvis | Hallux                | Big toe                      |
| Umbilical               | Navel or belly button             | Axillary              | Armpit                       |
| Pelvic                  | Lower abdomen or hip area         | Coxal                 | Hip                          |
| Femoral                 | Thigh                             |                       |                              |

## Posterior regions

### POSTERIOR REGIONAL TERMS

| Term             | Region                  | Term                       | Region                                 |
|------------------|-------------------------|----------------------------|----------------------------------------|
| <b>Cephalic</b>  | Head                    | <b>Sacral</b>              | Area between the hips, over the sacrum |
| <b>Occipital</b> | Back of the head        | <b>Gluteal</b>             | Buttock region                         |
| <b>Cervical</b>  | Neck                    | <b>Femoral</b>             | Thigh                                  |
| <b>Acromial</b>  | Point of the shoulder   | <b>Popliteal</b>           | Back of the knee                       |
| <b>Scapular</b>  | Shoulder blade          | <b>Sural</b>               | Calf                                   |
| <b>Dorsal</b>    | Back                    | <b>Fibular or peroneal</b> | Lateral side of the leg                |
| <b>Vertebral</b> | Spinal column           | <b>Calcaneal</b>           | Heel                                   |
| <b>Olecranal</b> | Back of the elbow       | <b>Plantar</b>             | Sole of the foot                       |
| <b>Lumbar</b>    | Lower back or loin area |                            |                                        |

2. **Identify those structures in terms of body orientation and direction.** Be able to use each in a sentence. The trachea is ventral to the spine. The nose is medial to the cheekbone.

### DIRECTIONAL TERMS, IN PAIRS

| Term                      | Its opposite       | What the pair describes                                |
|---------------------------|--------------------|--------------------------------------------------------|
| <b>Anterior (ventral)</b> | Posterior (dorsal) | Toward the front, versus toward the back               |
| <b>Superior</b>           | Inferior           | Above, versus below                                    |
| <b>Medial</b>             | Lateral            | Toward the midline, versus away from it                |
| <b>Cephalad (cranial)</b> | Caudal             | Toward the head, versus toward the tail end            |
| <b>Proximal</b>           | Distal             | Nearer the trunk, versus farther from it. Limbs only   |
| <b>Superficial</b>        | Deep               | Toward the body surface, versus away from it           |
| <b>Ipsilateral</b>        | Contralateral      | On the same side of the body, versus the opposite side |

3. **Body planes and sections.** Midline, median (midsagittal), parasagittal, coronal (frontal), transverse (horizontal or cross-sectional), oblique.
4. **Body cavities.** Ventral and dorsal, superior and inferior, cranial, vertebral, thoracic, abdominopelvic, mediastinum. What is the name of the structure that separates the thoracic cavity from the abdominopelvic cavity?

5. **Mediastinum.**

1. Know the structures that form the following borders: superior, inferior, ventral, dorsal, and lateral.

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2. Know the contents of the mediastinum.

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6. **Retroperitoneum.** Name the retroperitoneal structures. Memory device: SADPUCKER.

7. **Serous membranes and cavities.** Pericardial (pericardium): parietal, visceral, cavity, fluid. Peritoneal (peritoneum): parietal, visceral, cavity, fluid. Pleural (pleura): parietal, visceral, cavity, fluid.

8. **Abdominopelvic quadrants and the major structures found within each.**

#### THE FOUR QUADRANTS AND THEIR ORGANS

| Quadrant                 | Organs found within                                                                                                                                            |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Right upper (RUQ)</b> | Liver (right lobe), gallbladder, right kidney, duodenum, head of the pancreas, hepatic flexure of the colon, part of the ascending and transverse colon        |
| <b>Left upper (LUQ)</b>  | Stomach, spleen, left lobe of the liver, left kidney, body and tail of the pancreas, splenic flexure of the colon, part of the transverse and descending colon |
| <b>Right lower (RLQ)</b> | Cecum, appendix, ascending colon, right ureter, right ovary and uterine tube, right spermatic cord                                                             |
| <b>Left lower (LLQ)</b>  | Descending colon, sigmoid colon, left ureter, left ovary and uterine tube, left spermatic cord                                                                 |

The urinary bladder and the uterus sit at the midline in the hypogastric region and can be described in either lower quadrant.

9. **Be able to calculate total magnification, and name the parts of the microscope.**

## CHAPTER 2

# Epithelial Tissues

Identify the tissue on the slide, then write its function and location. Tick each structure as you find it.

### General structures, on every epithelial slide

#### STRUCTURES TO IDENTIFY

- |                                            |                                                       |                                                     |
|--------------------------------------------|-------------------------------------------------------|-----------------------------------------------------|
| <input type="checkbox"/> Apical surface    | <input type="checkbox"/> Lamina propria, when present | <input type="checkbox"/> Microvilli, when present   |
| <input type="checkbox"/> Lateral surface   | <input type="checkbox"/> Nuclei                       | <input type="checkbox"/> Goblet cells, when present |
| <input type="checkbox"/> Basal surface     | <input type="checkbox"/> Lumen or free surface        | <input type="checkbox"/> Keratin, when present      |
| <input type="checkbox"/> Basement membrane | <input type="checkbox"/> Cilia, when present          |                                                     |

#### ☐ Simple squamous

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

#### ☐ Keratinized stratified squamous

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

☐ **Non-keratinized stratified squamous**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

☐ **Simple cuboidal**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

☐ **Stratified cuboidal**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

☐ **Simple columnar**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

☐ **Pseudostratified ciliated columnar**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

☐ **Transitional**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

Which tissues typically contain cilia? \_\_\_\_\_ Function: \_\_\_\_\_

Microvilli? \_\_\_\_\_ Function: \_\_\_\_\_

Goblet cells? \_\_\_\_\_ Function: \_\_\_\_\_

**CHAPTER 3****Connective Tissues**

Connective tissue proper first, then cartilage and bone. For Module 1 you need the basic identification. The structures inside cartilage and bone come back in Module 2.

**Connective tissue proper**☐ **Loose areolar**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

**STRUCTURES TO IDENTIFY**☐ Collagen fibers☐ Fibroblasts☐ Elastic fibers☐ Mast cells☐ **Adipose**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

**STRUCTURES TO IDENTIFY**☐ Adipocytes☐ Fat vacuoles containing lipids☐ Blood vessels

☐ **Reticular**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

## STRUCTURES TO IDENTIFY

☐ Reticular fibers☐ **Dense regular**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

## STRUCTURES TO IDENTIFY

☐ Collagen fibers☐ Fibroblasts☐ **Dense irregular**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

## STRUCTURES TO IDENTIFY

☐ Collagen fibers☐ Fibroblasts☐ **Elastic**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

## STRUCTURES TO IDENTIFY

☐ Elastin fibers**Cartilage and osseous tissue** BASIC ID IN MODULE 1 STRUCTURES IN MODULE 2

Name the tissue, say what it does and where it is. Tick the structures you can already find, the rest is Module 2 work.

☐ **Hyaline cartilage**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

## STRUCTURES TO IDENTIFY

☐ Perichondrium☐ Chondrocytes☐ Lacunae☐ Extracellular matrix☐ **Elastic cartilage**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

## STRUCTURES TO IDENTIFY

☐ Perichondrium☐ Chondrocytes☐ Elastin fibers☐ Lacunae☐ Extracellular matrix☐ **Fibrocartilage**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

## STRUCTURES TO IDENTIFY

☐ Lacunae☐ Extracellular matrix☐ No perichondrium☐ Chondrocytes☐ Collagen fibers

☐ **Osseous tissue**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

## STRUCTURES TO IDENTIFY

- |                                                    |                                   |                                     |
|----------------------------------------------------|-----------------------------------|-------------------------------------|
| <input type="checkbox"/> Osteons                   | <input type="checkbox"/> Lacunae  | <input type="checkbox"/> Osteocytes |
| <input type="checkbox"/> Central (Haversian) canal | <input type="checkbox"/> Lamellae | <input type="checkbox"/> Canaliculi |

## CHAPTER 4

**Muscle Tissues**

Basic identification for Module 1. Name the tissue, what it does, where it is, and whether it is voluntary or involuntary. The sarcomere and the connective tissue wrappings come back in Module 3.

☐ **Skeletal muscle**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

VOLUNTARY OR INVOLUNTARY \_\_\_\_\_

## STRUCTURES TO IDENTIFY

- |                                                     |                                                          |                                     |
|-----------------------------------------------------|----------------------------------------------------------|-------------------------------------|
| <input type="checkbox"/> Striations, dark and light | <input type="checkbox"/> Nuclei, multiple and peripheral | <input type="checkbox"/> Sarcolemma |
|-----------------------------------------------------|----------------------------------------------------------|-------------------------------------|

☐ **Cardiac muscle**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

VOLUNTARY OR INVOLUNTARY \_\_\_\_\_

## STRUCTURES TO IDENTIFY

- |                                             |                                                 |
|---------------------------------------------|-------------------------------------------------|
| <input type="checkbox"/> Branching fibers   | <input type="checkbox"/> Single central nucleus |
| <input type="checkbox"/> Intercalated discs | <input type="checkbox"/> Striations             |

☐ **Smooth muscle**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

VOLUNTARY OR INVOLUNTARY \_\_\_\_\_

## STRUCTURES TO IDENTIFY

- |                                               |                                                 |                                        |
|-----------------------------------------------|-------------------------------------------------|----------------------------------------|
| <input type="checkbox"/> Spindle-shaped cells | <input type="checkbox"/> Single central nucleus | <input type="checkbox"/> No striations |
|-----------------------------------------------|-------------------------------------------------|----------------------------------------|

What two cell junctions are found at intercalated discs, and what does each one do?

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## CHAPTER 5

**Nerve Tissues**

Basic identification for Module 1. The detail comes back in Module 5.

☐ **Multipolar neuron**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

## STRUCTURES TO IDENTIFY

- |                                           |                                       |                                        |
|-------------------------------------------|---------------------------------------|----------------------------------------|
| <input type="checkbox"/> Cell body (soma) | <input type="checkbox"/> Dendrites    | <input type="checkbox"/> Synaptic bulb |
| <input type="checkbox"/> Nucleus          | <input type="checkbox"/> Axon         |                                        |
| <input type="checkbox"/> Nucleolus        | <input type="checkbox"/> Axon hillock |                                        |

☐ **Spinal cord**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

## STRUCTURES TO IDENTIFY

- |                                       |                                       |                                                     |
|---------------------------------------|---------------------------------------|-----------------------------------------------------|
| <input type="checkbox"/> White matter | <input type="checkbox"/> Ventral horn | <input type="checkbox"/> Lateral horn, when present |
| <input type="checkbox"/> Grey matter  | <input type="checkbox"/> Dorsal horn  |                                                     |

☐ **Cerebral cortex**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

## STRUCTURES TO IDENTIFY

- ☐
- Pyramidal cells

☐ **Cerebellum**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

## STRUCTURES TO IDENTIFY

- ☐
- Purkinje cells

☐ **Nerve, cross section**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

## STRUCTURES TO IDENTIFY

- |                                        |                                           |                                      |
|----------------------------------------|-------------------------------------------|--------------------------------------|
| <input type="checkbox"/> Axon          | <input type="checkbox"/> Nodes of Ranvier | <input type="checkbox"/> Perineurium |
| <input type="checkbox"/> Schwann cells | <input type="checkbox"/> Epineurium       | <input type="checkbox"/> Endoneurium |

## CHAPTER 6

**Blood Vessels and Formed Elements**

Basic identification for Module 1. The detail comes back in Module 3.

**Vessel walls**☐ **Artery**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

## STRUCTURES TO IDENTIFY

- |                                                  |                                                          |                                         |
|--------------------------------------------------|----------------------------------------------------------|-----------------------------------------|
| <input type="checkbox"/> Tunica intima           | <input type="checkbox"/> Tunica media                    | <input type="checkbox"/> Tunica externa |
| <input type="checkbox"/> Endothelium             | <input type="checkbox"/> Smooth muscle                   | <input type="checkbox"/> Lumen          |
| <input type="checkbox"/> Internal elastic lamina | <input type="checkbox"/> Dense elastic connective tissue | <input type="checkbox"/> Erythrocytes   |

☐ **Vein**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

## STRUCTURES TO IDENTIFY

- ☐ Tunica intima  
☐ Endothelium

- ☐ Tunica media with smooth muscle  
☐ Tunica externa

- ☐ Lumen  
☐ Erythrocytes

**Peripheral blood smear**

Identify each formed element, then write its function and how you recognized it.

☐ **Neutrophil**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

☐ **Basophil**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

☐ **Eosinophil**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

☐ **Monocyte**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

☐ **Lymphocyte**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

☐ **Erythrocyte**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

☐ **Thrombocyte**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

## CHAPTER 7

**Integumentary System**

Layers in order, deep to superficial, then the structures in the dermis and hypodermis.

**Strata of the epidermis**☐ **Stratum corneum**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_



☐ **Stratum lucidum**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

☐ **Stratum granulosum**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

☐ **Stratum spinosum**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

☐ **Stratum basale**

FUNCTION \_\_\_\_\_ LOCATION \_\_\_\_\_

**Dermis**☐ **Dermis, on slides and models**

## STRUCTURES TO IDENTIFY

- |                                                            |                                               |                                                   |
|------------------------------------------------------------|-----------------------------------------------|---------------------------------------------------|
| <input type="checkbox"/> Papillary layer                   | <input type="checkbox"/> Hair                 | <input type="checkbox"/> Sebaceous gland and duct |
| <input type="checkbox"/> Reticular layer                   | <input type="checkbox"/> Hair follicle        | <input type="checkbox"/> Apocrine gland and duct  |
| <input type="checkbox"/> Areolar connective tissue         | <input type="checkbox"/> Hair root            | <input type="checkbox"/> Meissner's corpuscles    |
| <input type="checkbox"/> Dense irregular connective tissue | <input type="checkbox"/> Hair bulb            | <input type="checkbox"/> Free nerve endings       |
| <input type="checkbox"/> Dermal papillae                   | <input type="checkbox"/> Hair shaft           | <input type="checkbox"/> Capillary loops          |
| <input type="checkbox"/> Eccrine sweat gland and duct      | <input type="checkbox"/> Arrector pili muscle |                                                   |

**Hypodermis, the subcutaneous tissue**☐ **Hypodermis**

## STRUCTURES TO IDENTIFY

- |                                                    |                                                    |                                                          |
|----------------------------------------------------|----------------------------------------------------|----------------------------------------------------------|
| <input type="checkbox"/> Adipose connective tissue | <input type="checkbox"/> Areolar connective tissue | <input type="checkbox"/> Lamellated (Pacinian) corpuscle |
|----------------------------------------------------|----------------------------------------------------|----------------------------------------------------------|

Name the three structures found within dermal papillae.

---

Which epidermal layer is missing in thin skin? \_\_\_\_\_